Foundations of Data Science for Biologists

Data visualization: graphical excellence and failure

BIO 724D

2025-SEP-30

Instructors: Greg Wray and Kayla Wilhoit

A historical tour of graphical excellence

The earliest known abstract graphic? (right)

Chauvet-Pont-d'Arc Cave, France; unknown artists; 32,000 - 30,000 BCE

First true map?

town and erupting volcano

Çatalhöyük, Turkey;

unknown designer; ~6200 BCE

YBC 7289

Geometric diagram;
derives square root of 2

1;24,51,10 base 60 or
305,470/216,000 = 1.4142129 decimal;
a second number is the reciprocal

Babylon;
unknown designer;
1800-1600 BCE

Plimpton 322

New York Times

Earliest known “table”;
multiplication base 60

Babylon;
unknown designer;
~1800 BCE

Earliest known "world map"

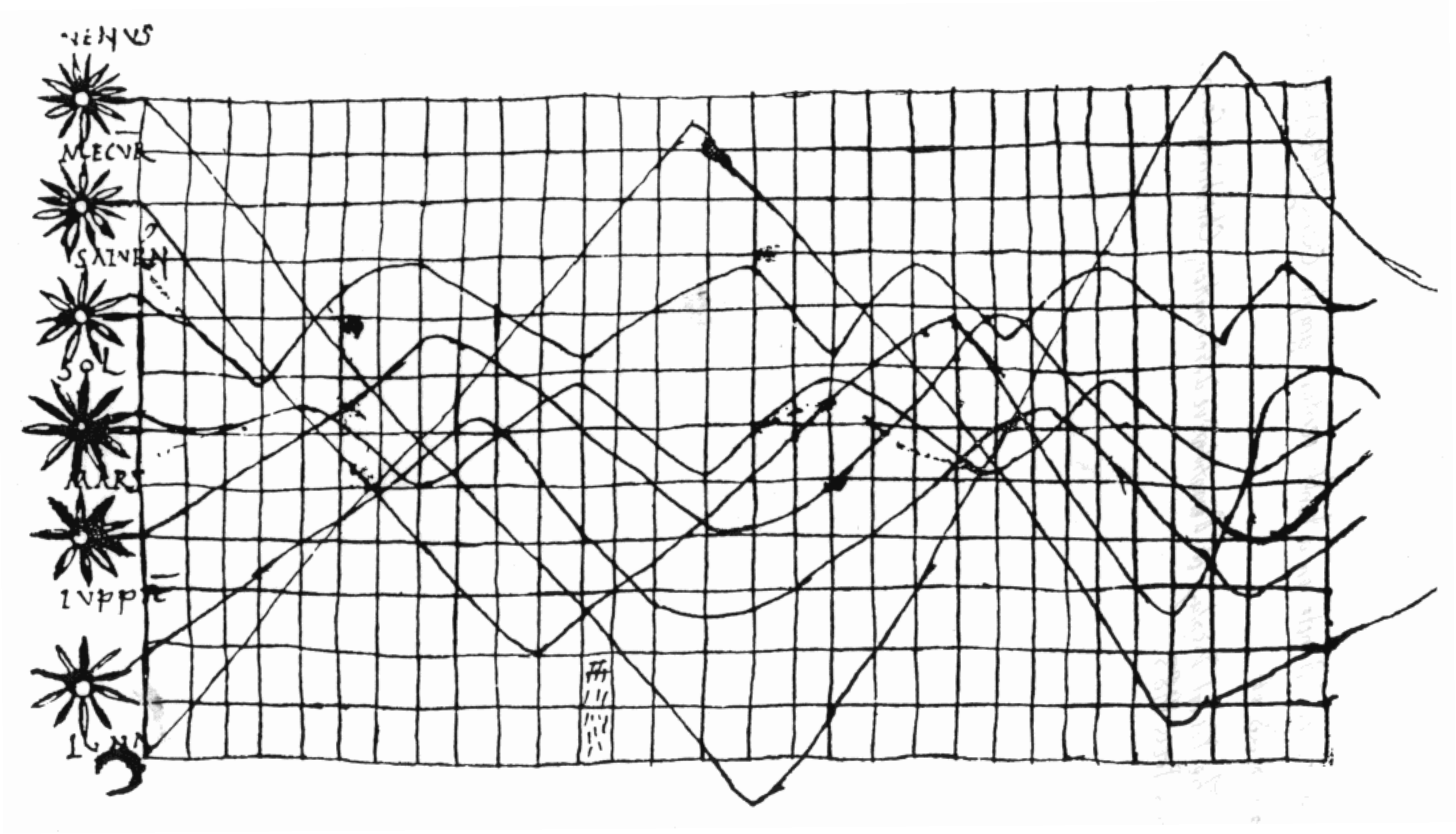
Babylon;
unknown designer;
~625 BCE

British Museum

First graphical representation of star positions;

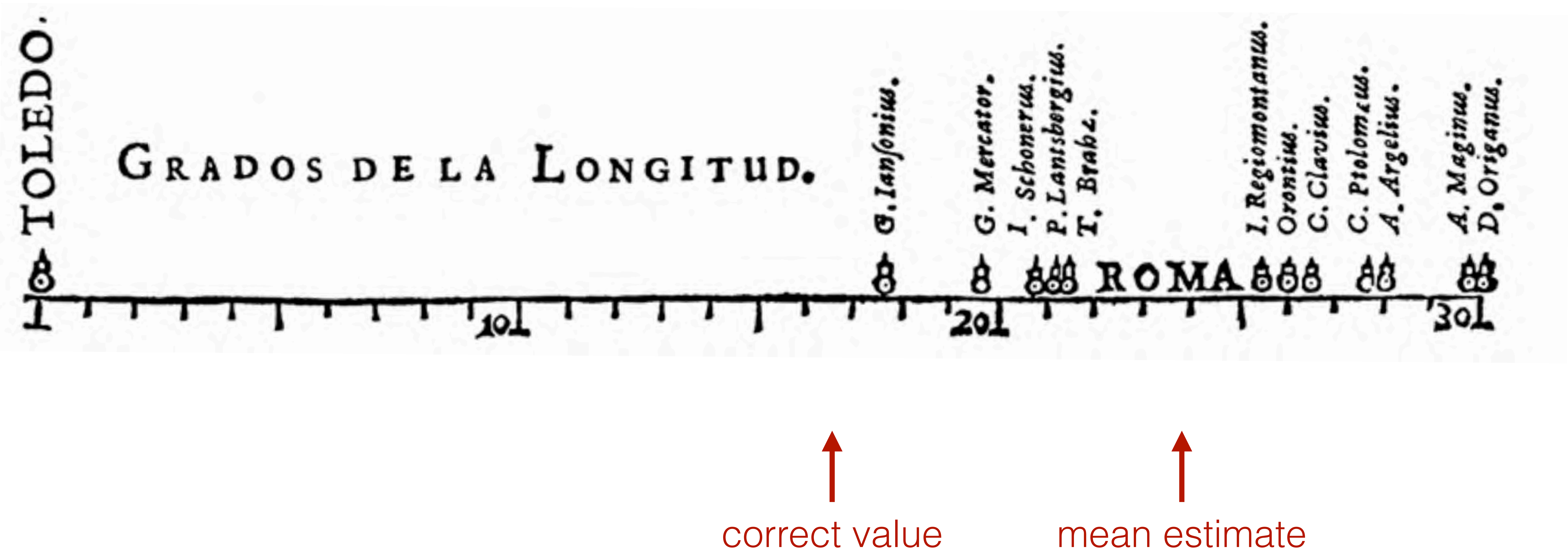
China;

unknown designer; 7th century



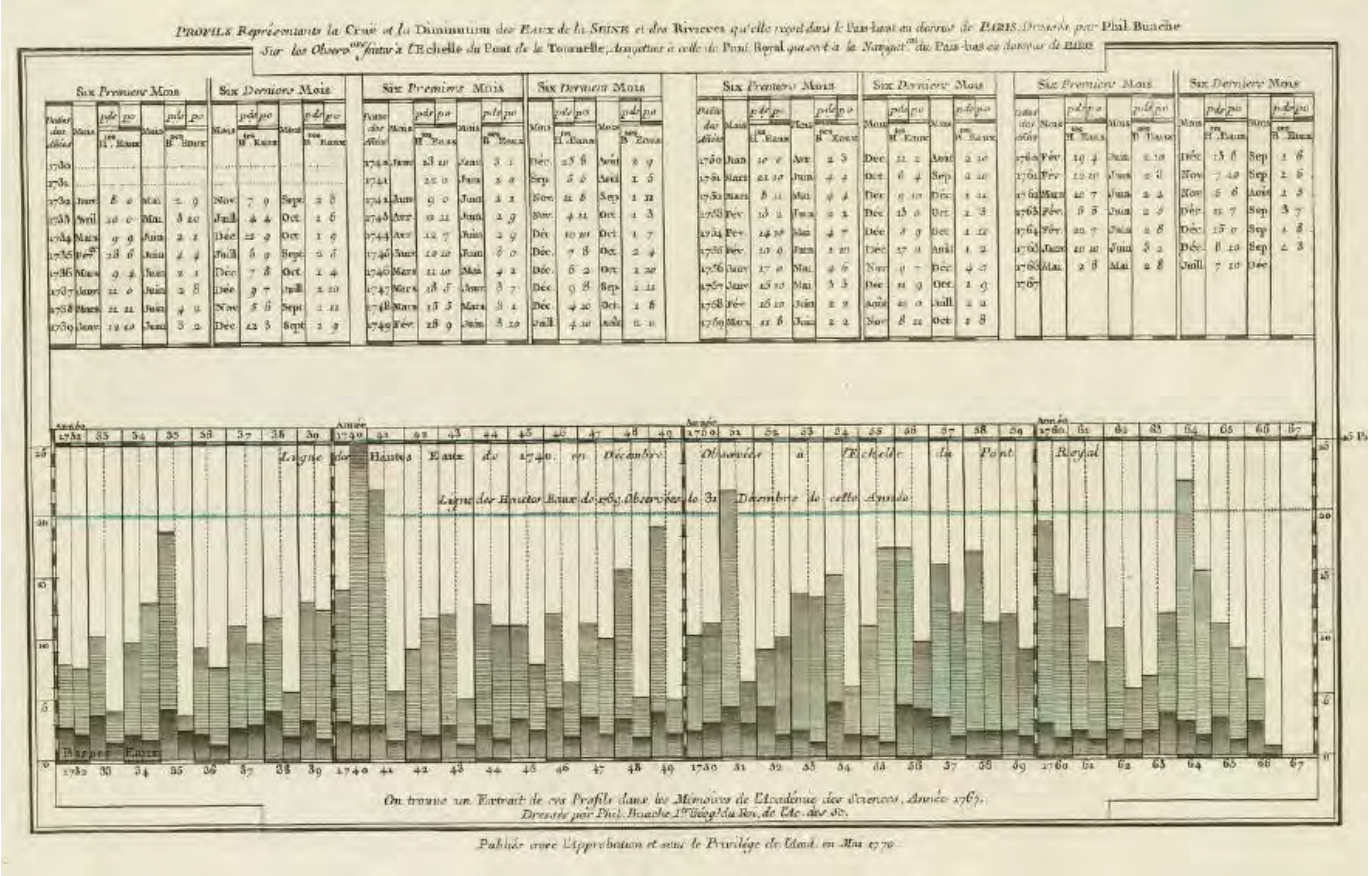
Earliest known line graph; annotated as *De cursu per zodiacum*

Unknown designer; early 11th century



Earliest known statistical graphic; estimates of degrees longitude Toledo-to-Rome

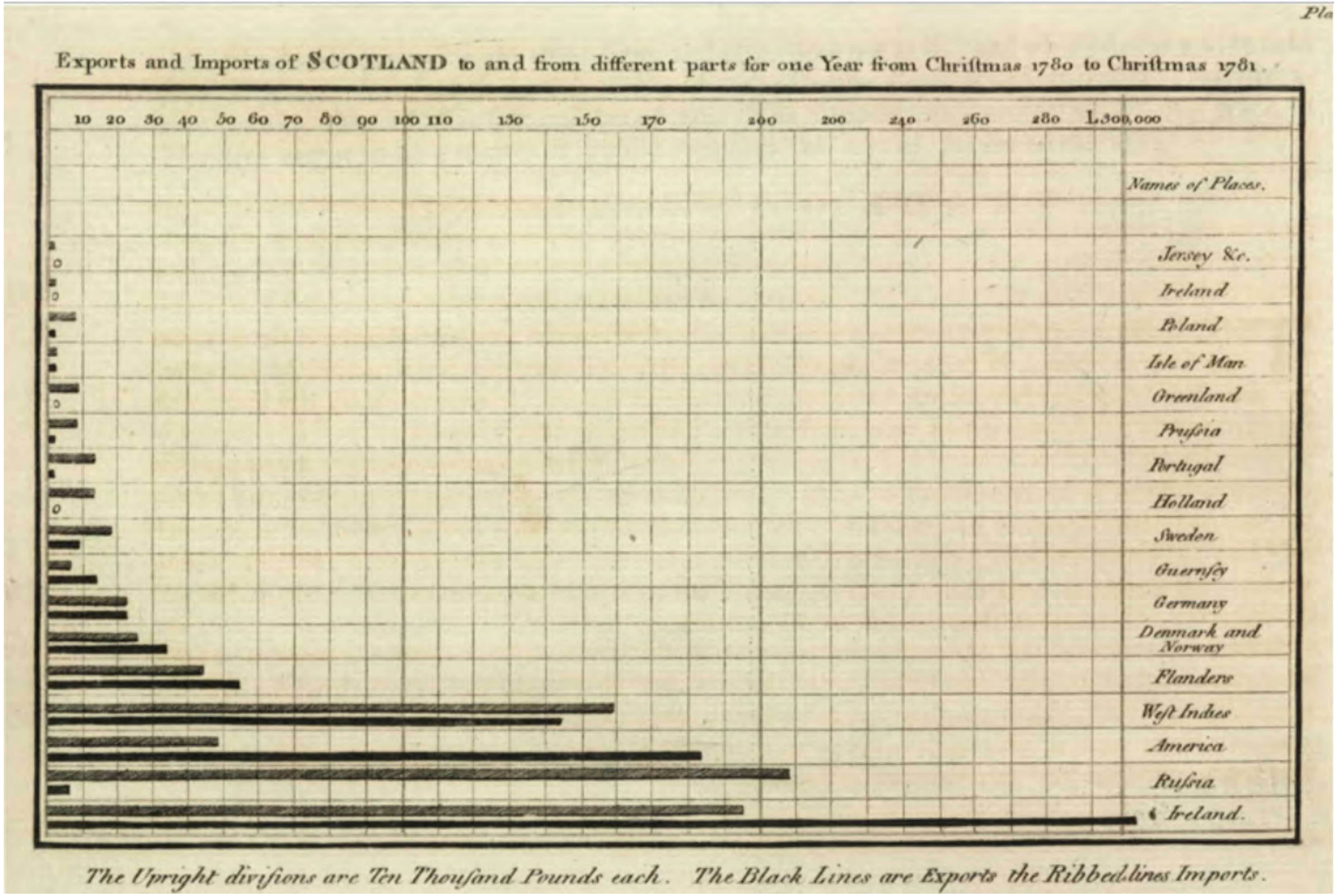
Michael Florent van Langren; 1644 (two earlier unpublished versions are known)



First known bar chart;
first time series chart;
categorical encoding
by shading bars

High and low water
marks of the Seine
in each year

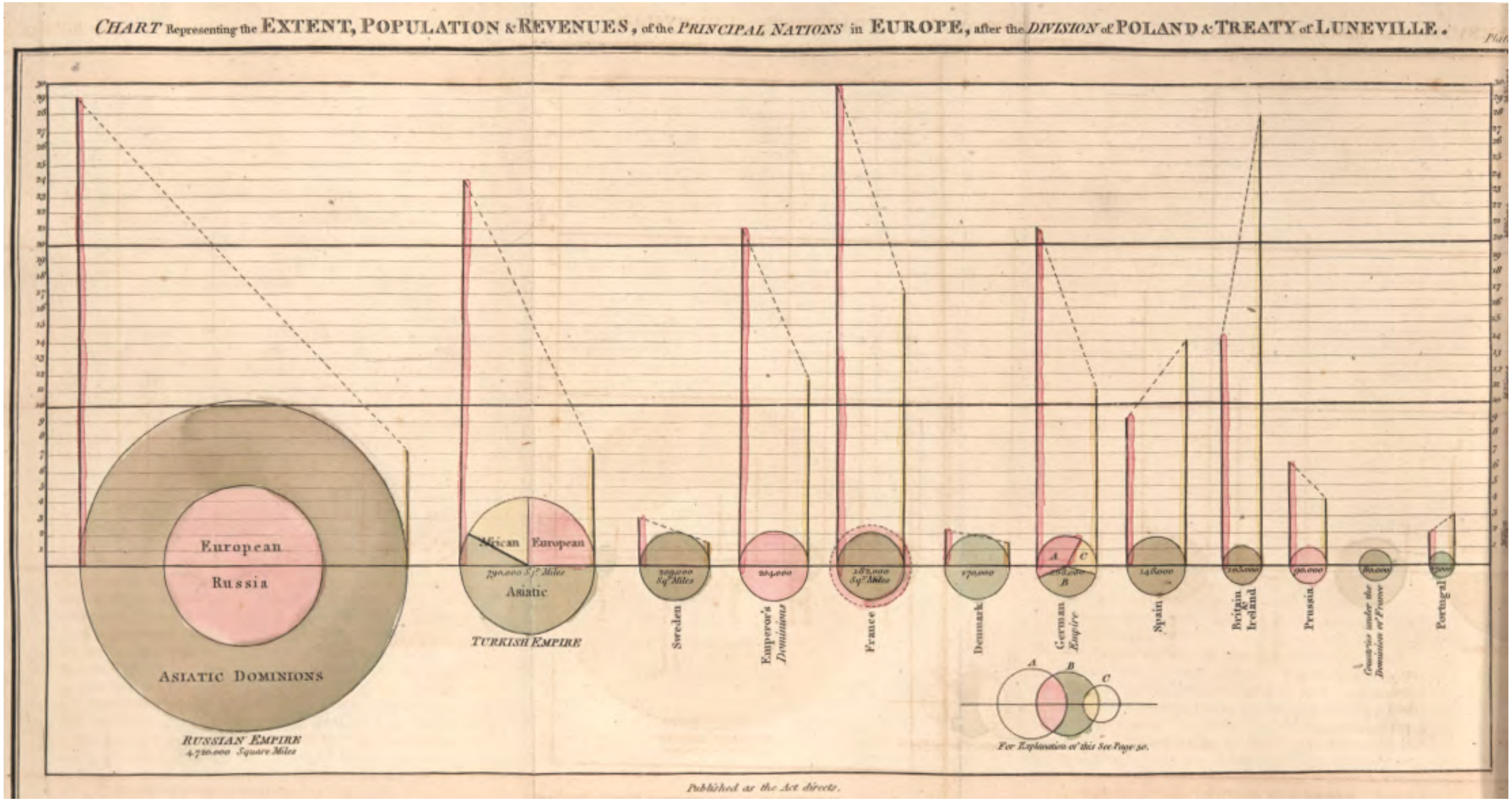
Philippe Bauche and
Guillame de L'Isle; 1754



Another early bar graph; categorical encoding

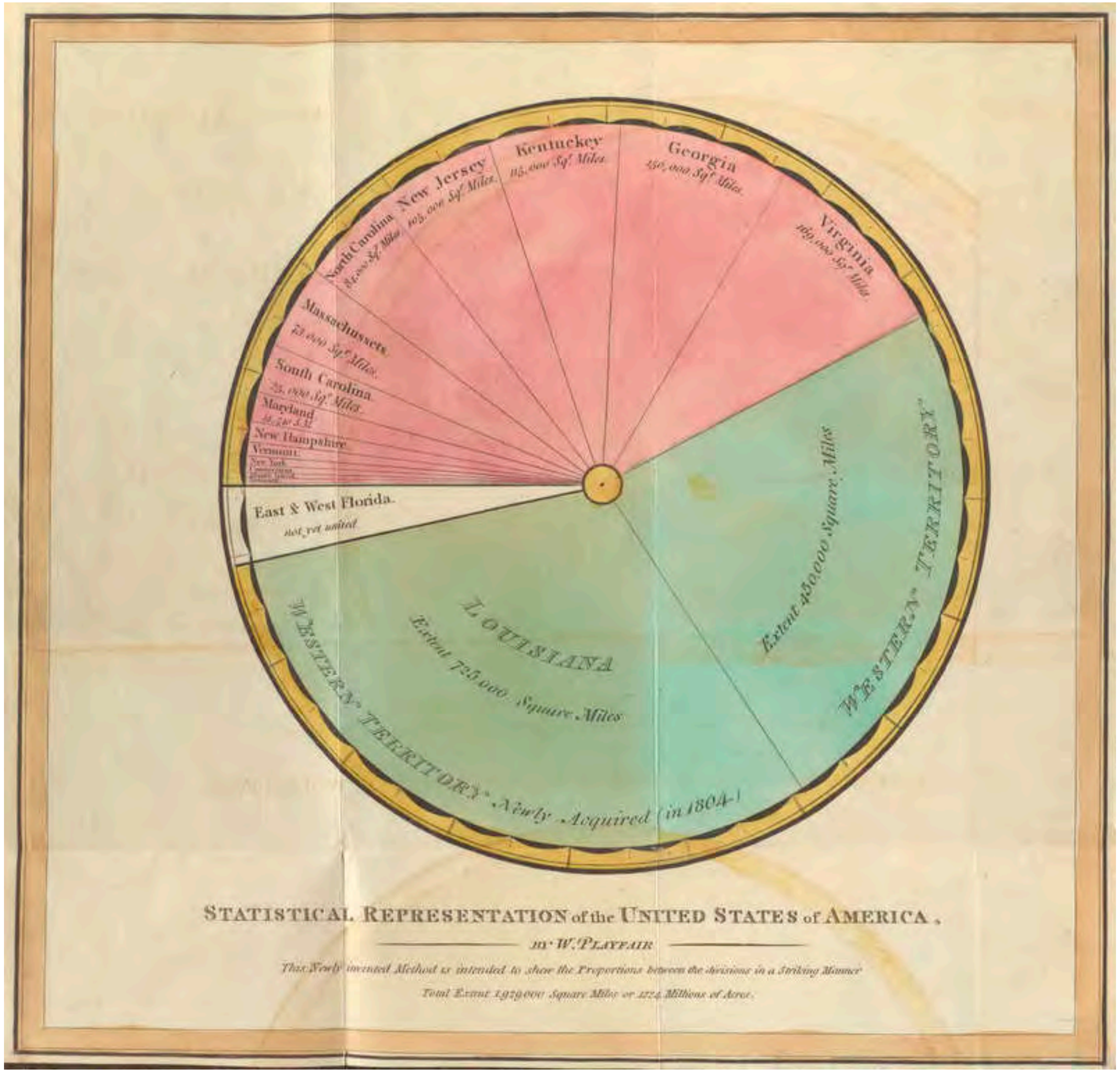
Imports and exports to different countries

William Playfair; 1781



Circles and bars to compare two distinct values; includes the first pie chart (Turkish Empire)

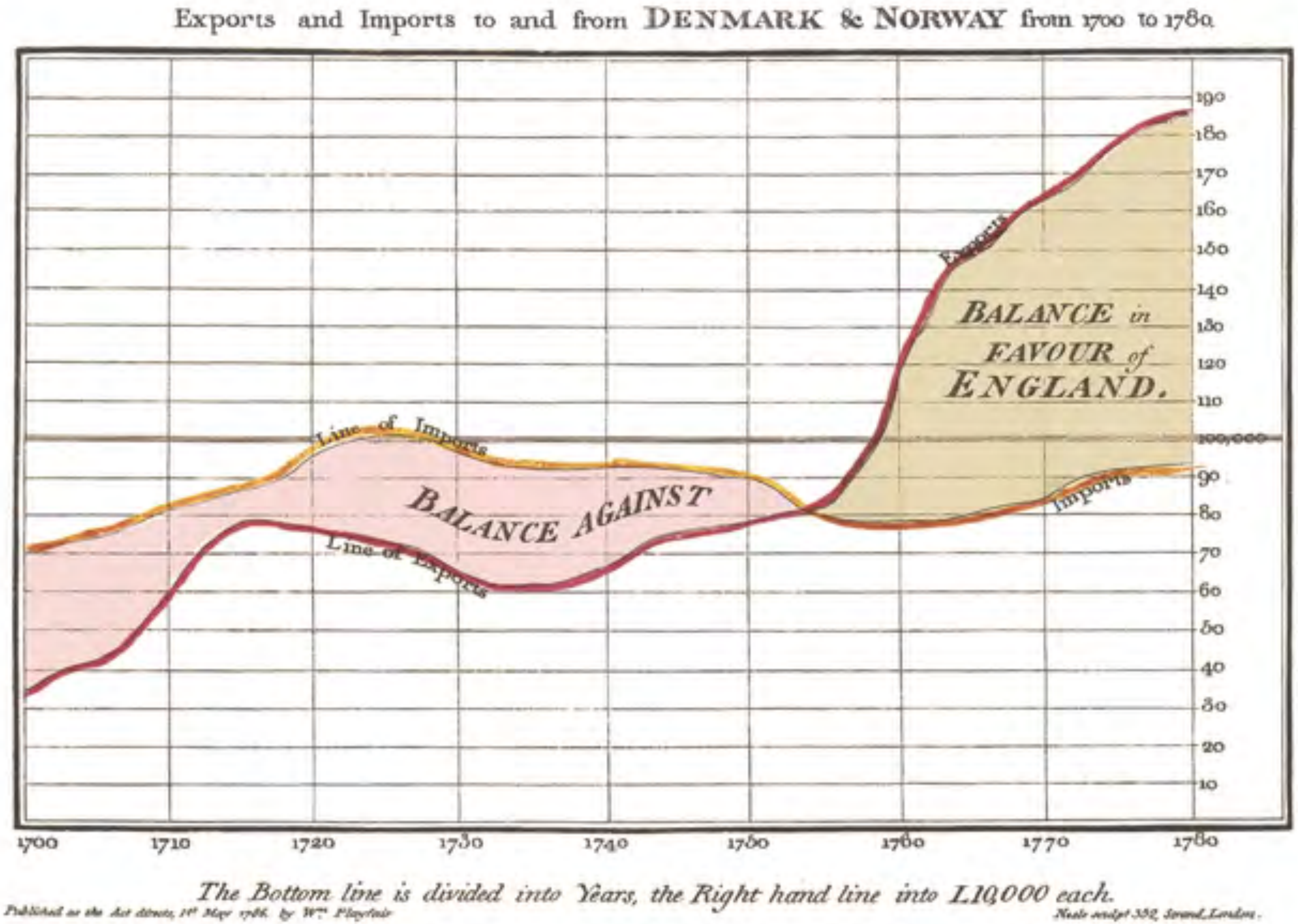
William Playfair; 1789



An early pie chart;
uses color to group values

Area of states and territories
in the USA c1800

William Playfair; 1805



Line charts that

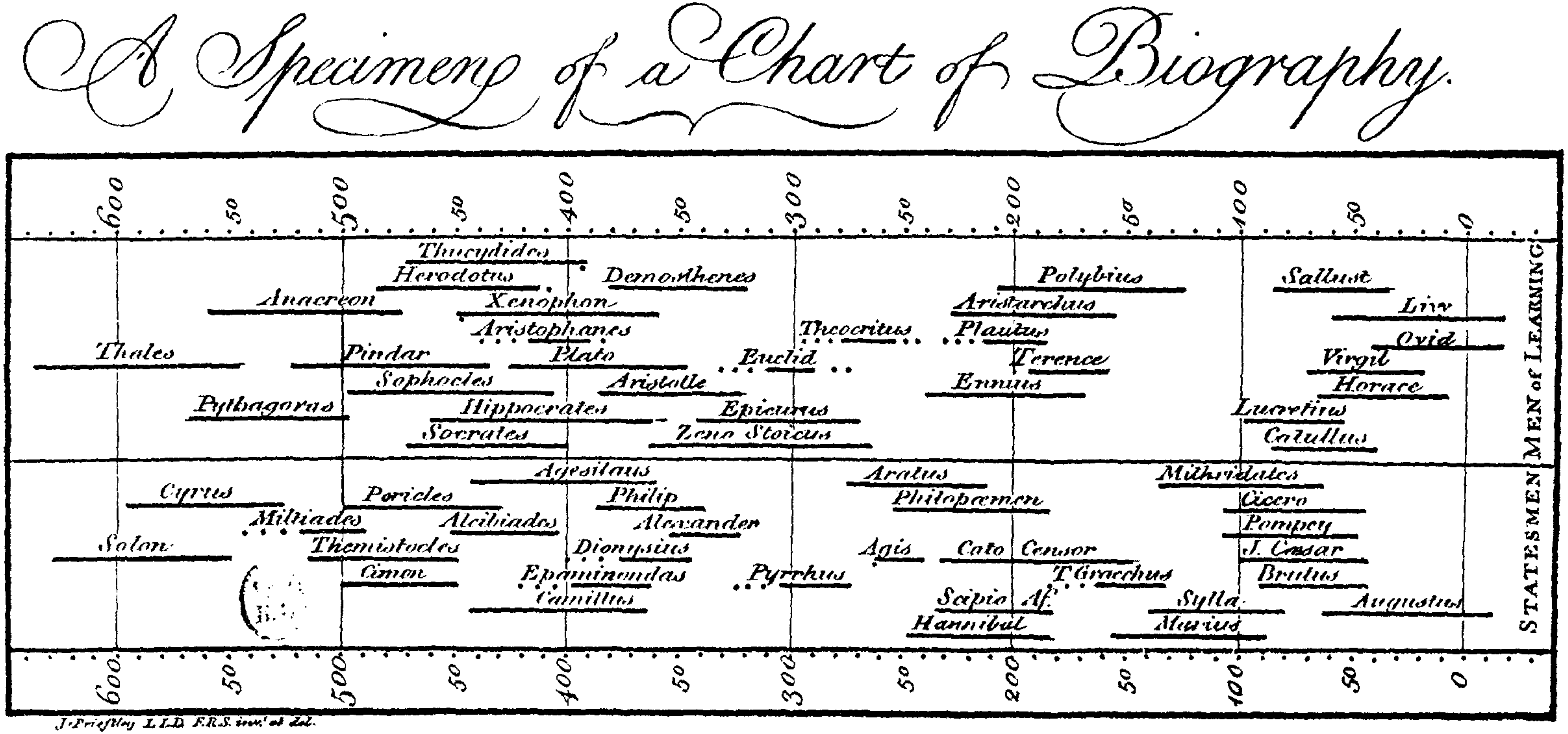
tell a story using

annotation and color

Imports and exports

over time

William Playfair; 1786



Invention of time-lines to compare intervals

Joseph Priestly; 1765

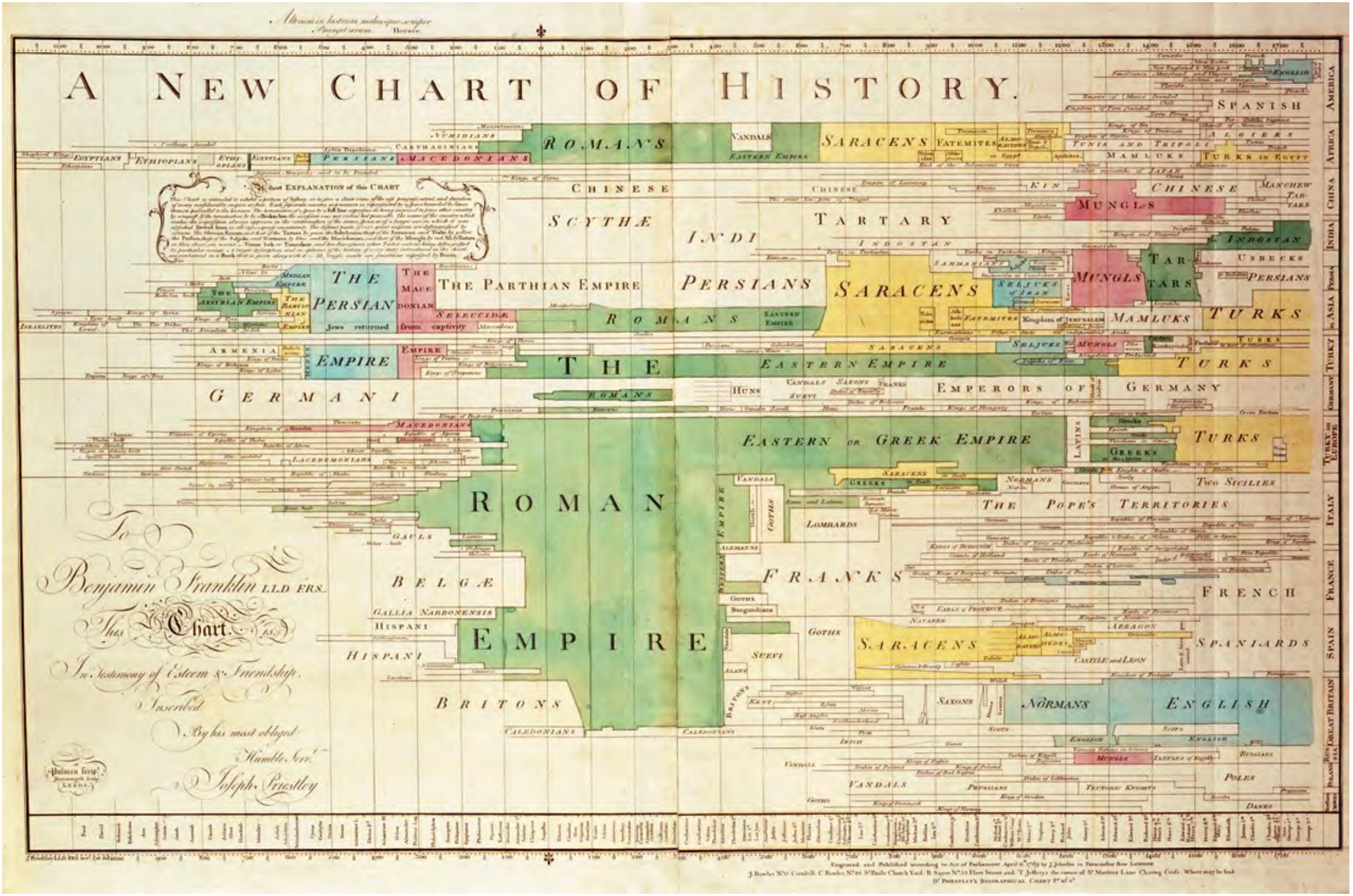
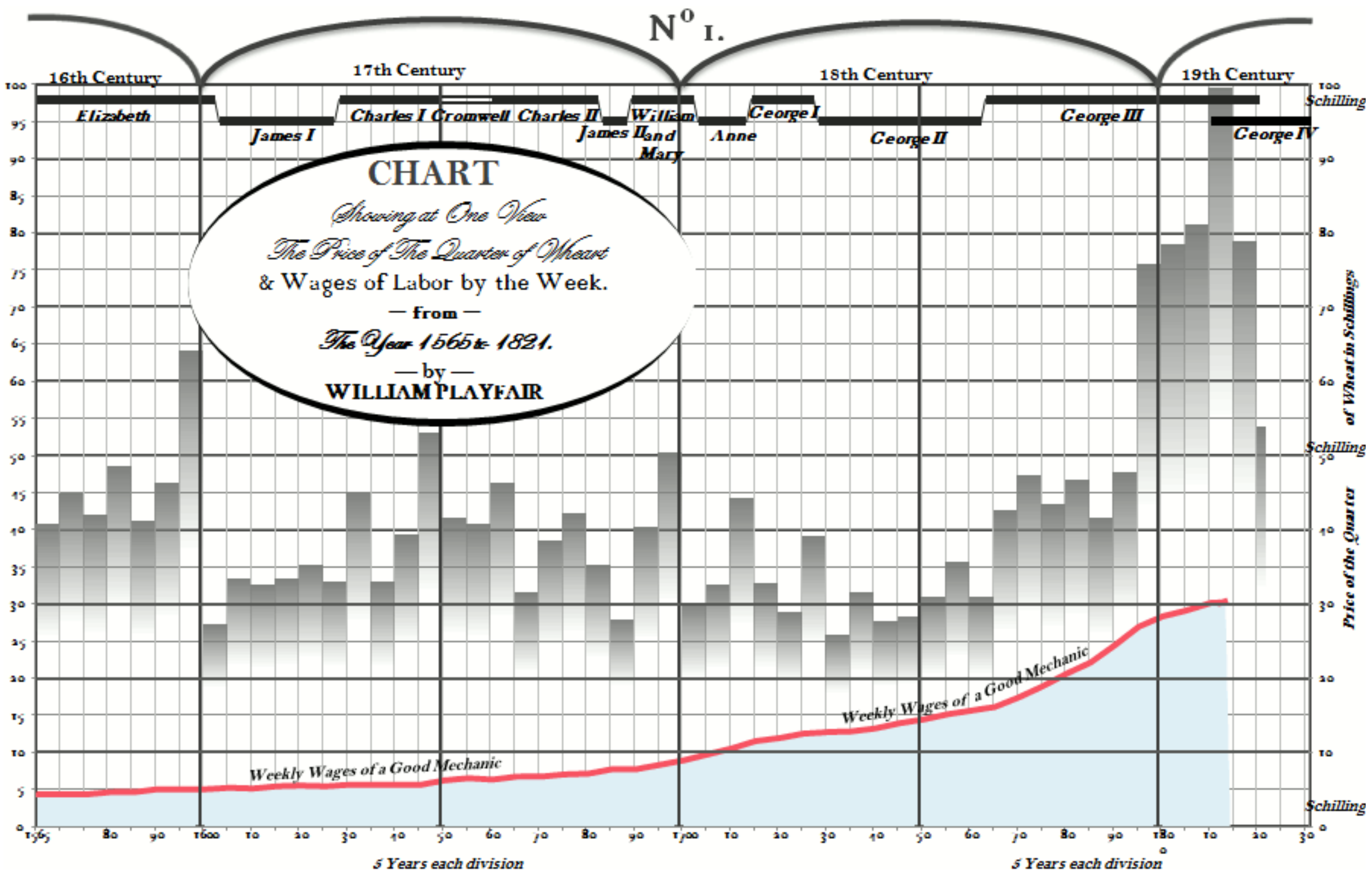
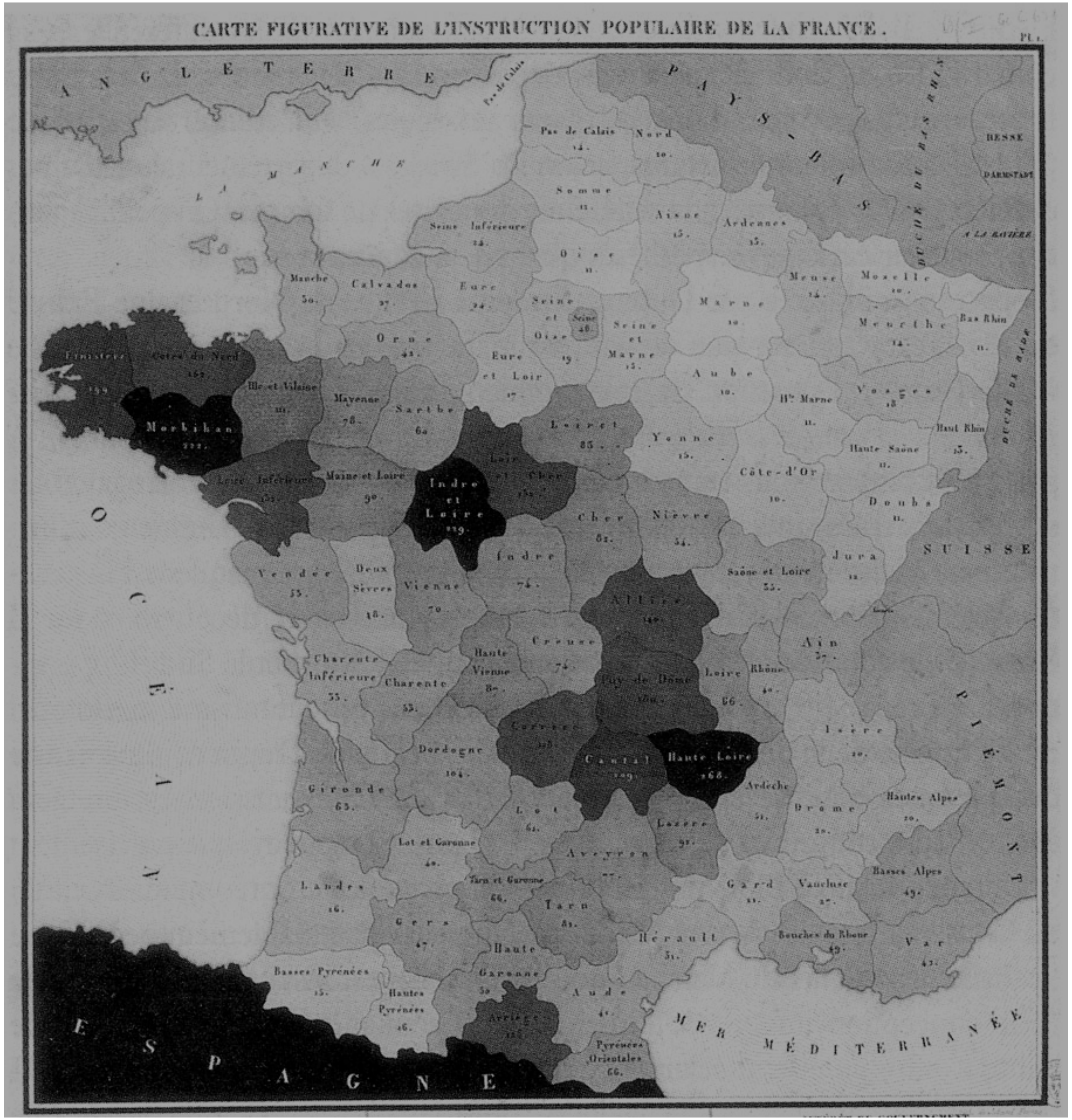


Chart of World History; use of color to encode a categorical variable; Joseph Priestly; 1765



Multivariate time series plots with time lines introduced by Joseph Priestly and William Playfair

William Playfair; 1821

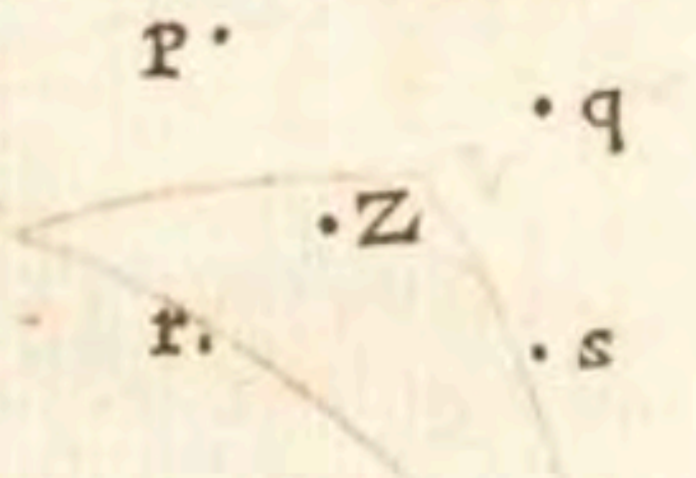


First choropleth map;

quantitative data and geography

Charles Dupin; 1826

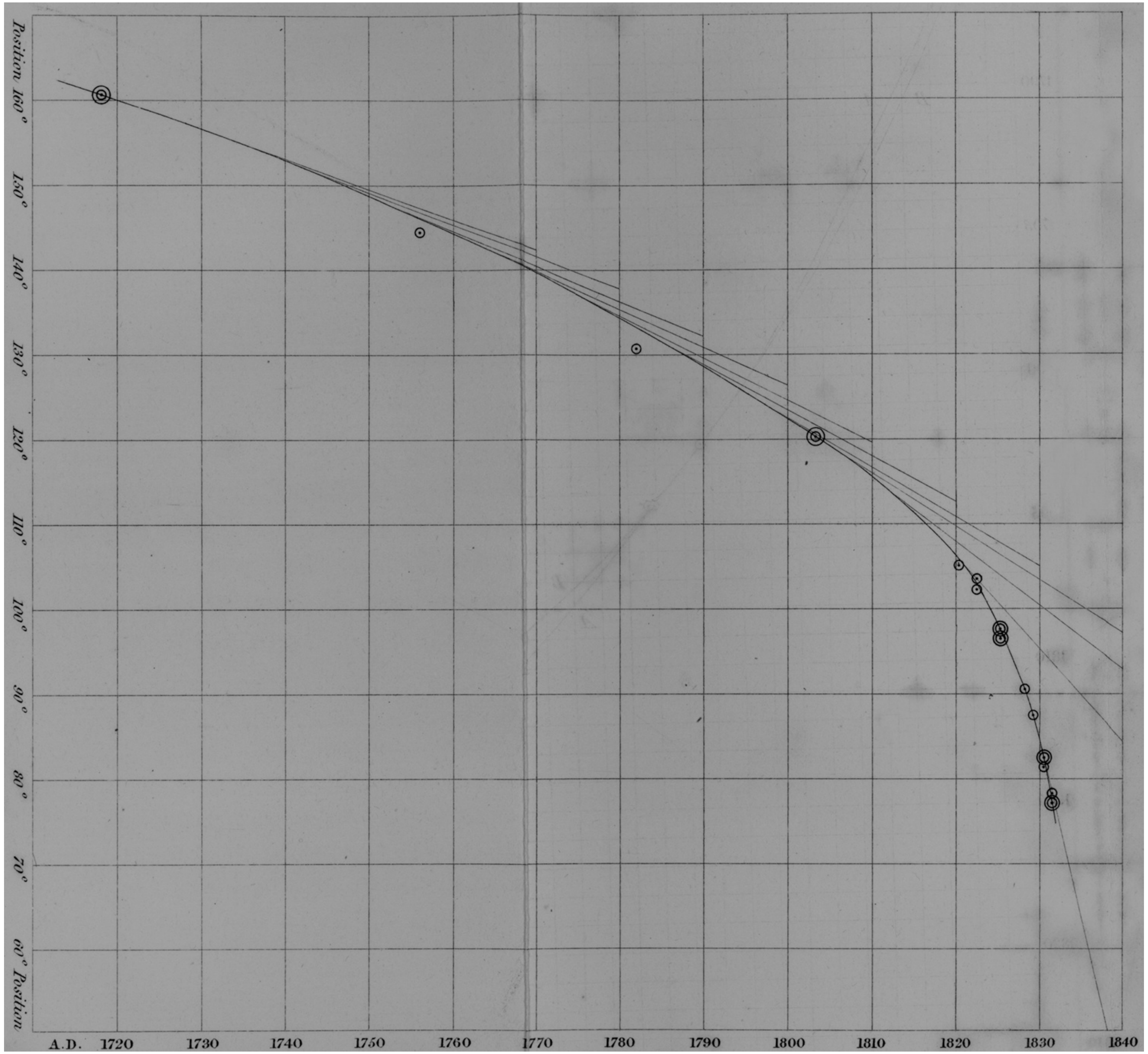
Ad eundem fere modum in aliis caſibus Limites inveniuntur Errorum qui ex minus accuratis obſervationibus ortum ducunt, quin & Poſitiones ad Obſervandum commodiſſimæ deprehenduntur: ut mihi vix quidquam ulterius deſiderari videatur poſtquam oſtenſum fuerit qua ratione Probabilitas maxima in his rebus haberi poſſit, ubi diverſæ Obſervationes, in eundem finem inſtitutæ, paullulum diverſas ab invicem concluſiones exhibent. Id autem fiet ad modum ſequentis Exempli. Sit p locus Objecti alicujus ex Obſervatione prima definitus, q, r, s ejuſdem Objecti loca ex Obſervationibus ſubſequentibus; ſint inſuper $\mathcal{P}$, $\mathcal{Q}$, R, S pondera reciproce proportionalia ſpatiis Evagationum, per quæ ſe diffundere poſſint Errores ex Obſervationibus ſingulis prodeuntes, quæque dantur ex datis Errorum Limitibus; & ad puncta p, q, r, s poſita intelligantur pondera $\mathcal{P}$, $\mathcal{Q}$, R, S, & inveniatur eorum gravitatis centrum Z: dico punctum Z fore Locum Objecti maxime probabilem, qui pro vero ejus loco tutiſſime haberi poteſt.



Almost a bivariate plot!

Graphical illustration of the concept of a weighted mean

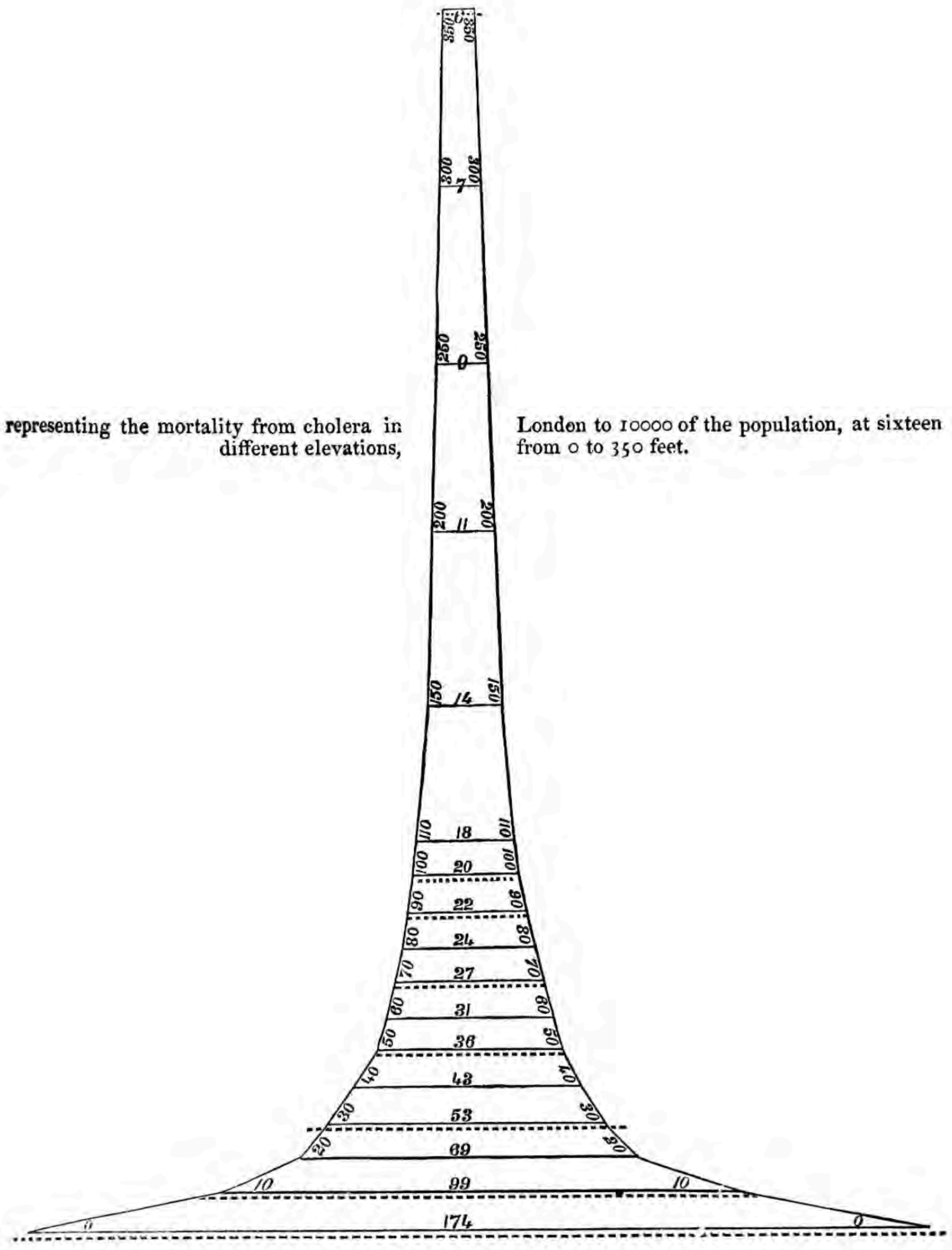
Roger Cotes; 1722



The first scatterplot;
relationship between variables

Positional angle of double stars
in relation to year of observation

John F. W. Herschel; 1833



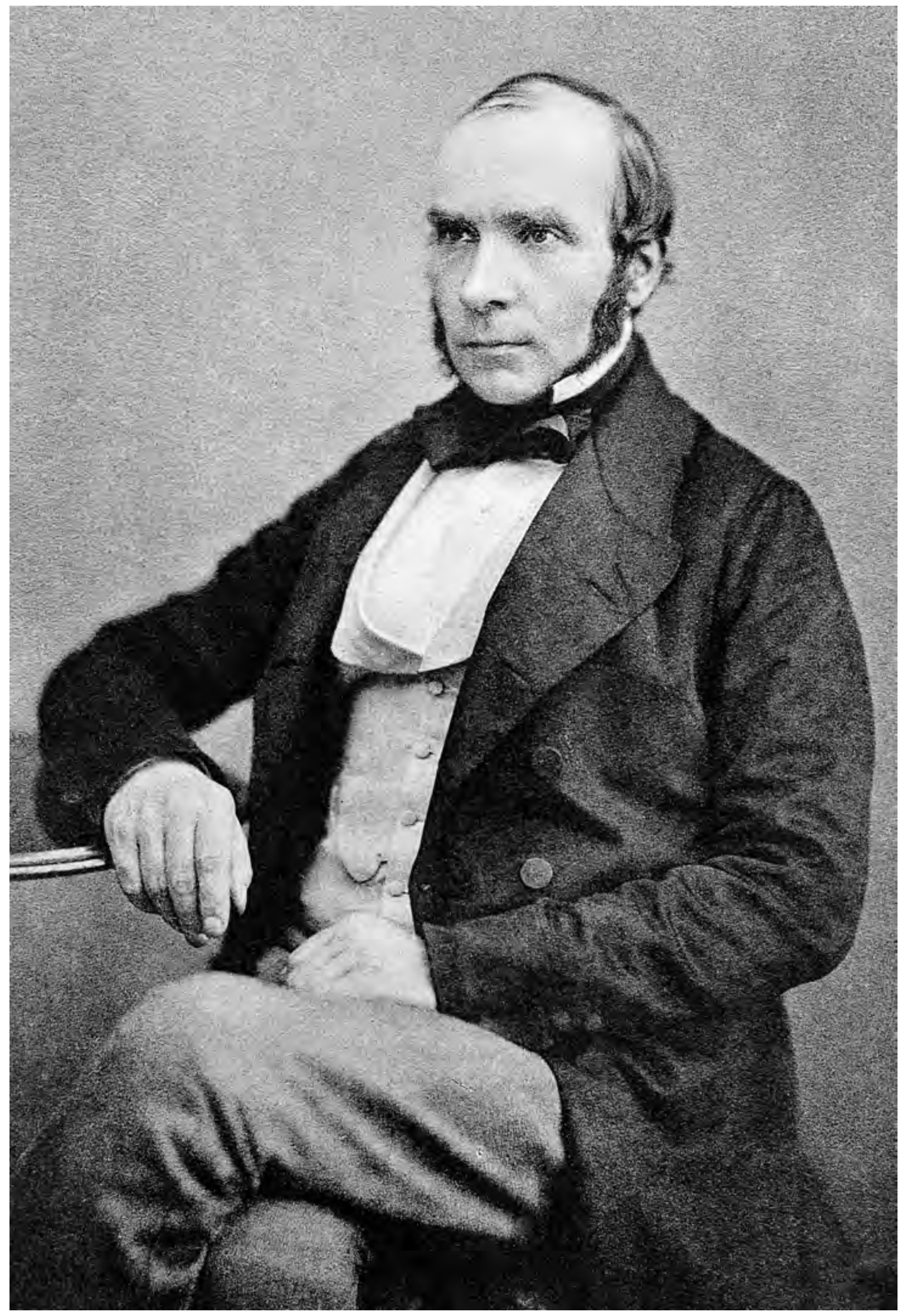
Relationship between two variables;
not as readable as a scatter plot

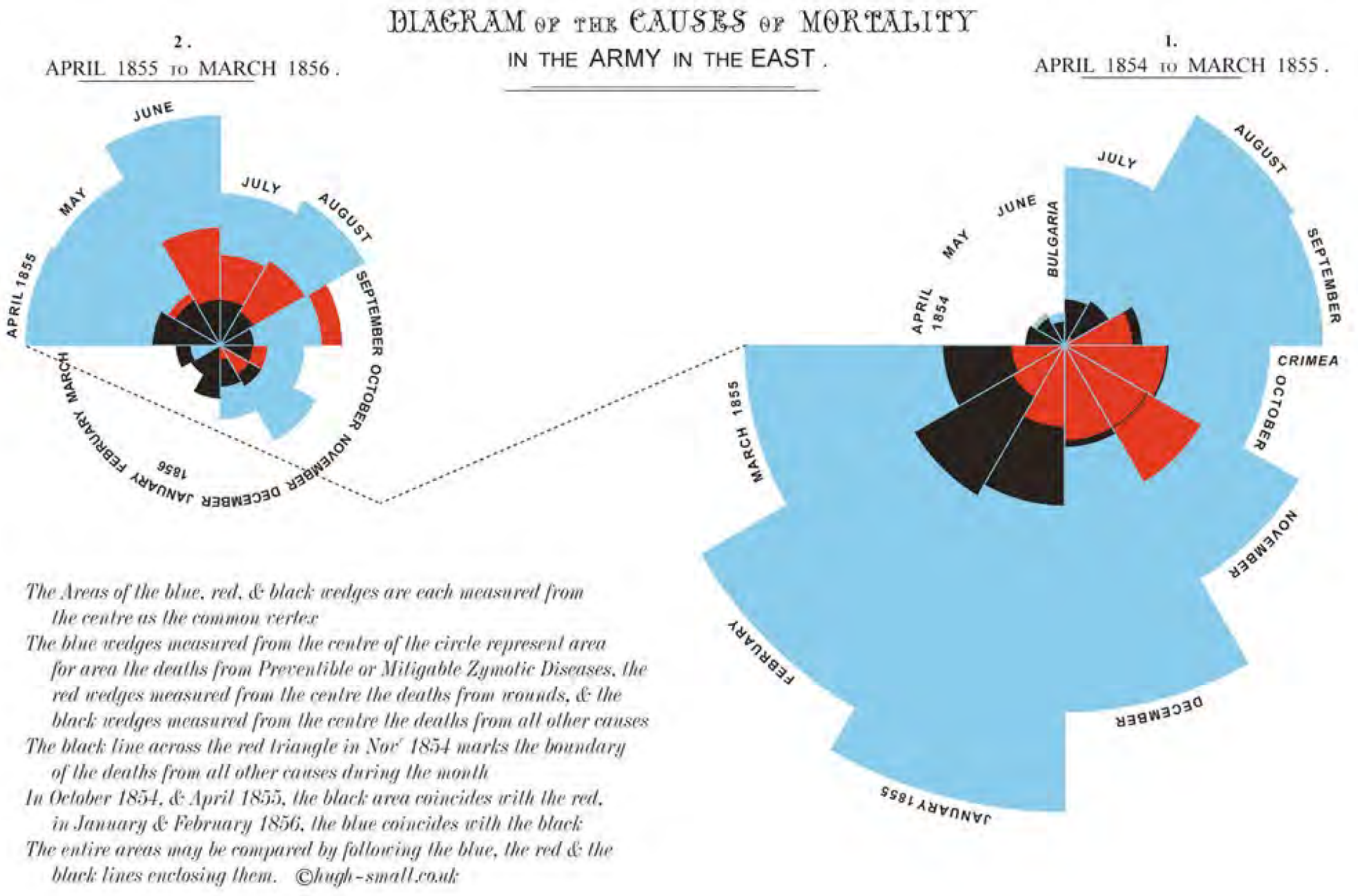
Inverse relationship between
mortality and elevation above
the Thames River

General Register Office; 1852

The birth of epidemiology; spatial plots

John Snow; 1854



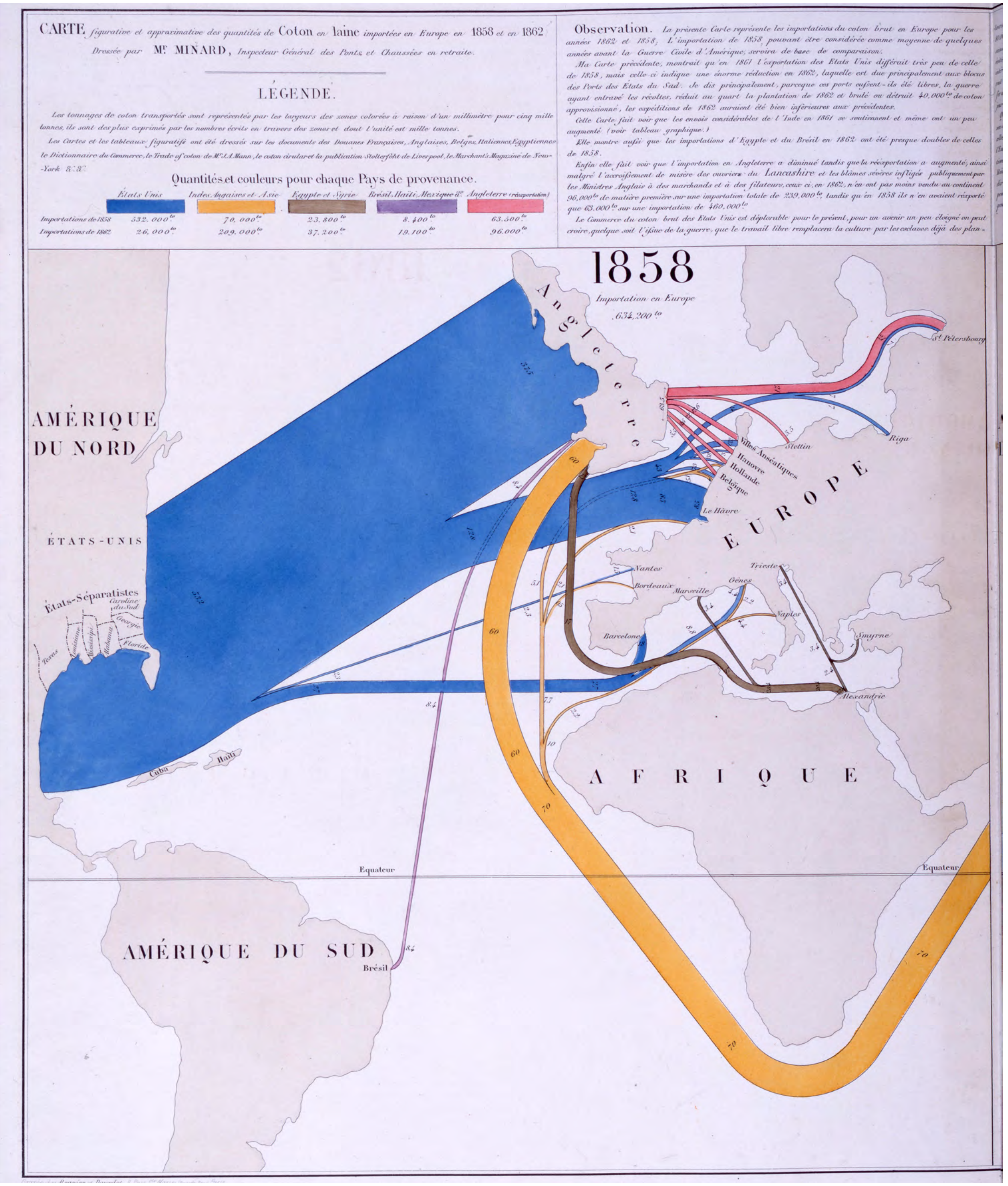


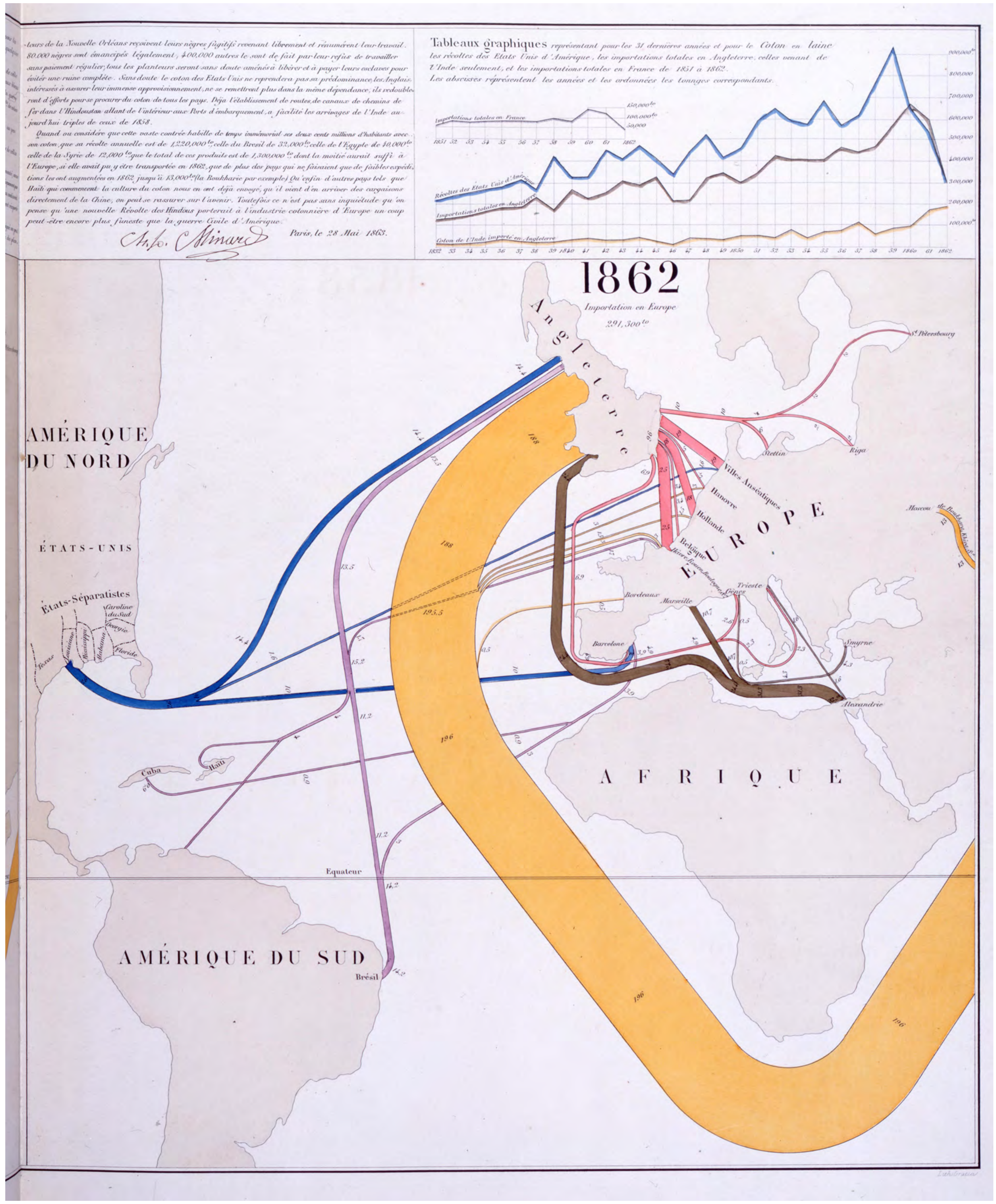
Polar area diagram;
use of color and proportion

Florence Nightingale; 1858

Causes of death during the Crimean war

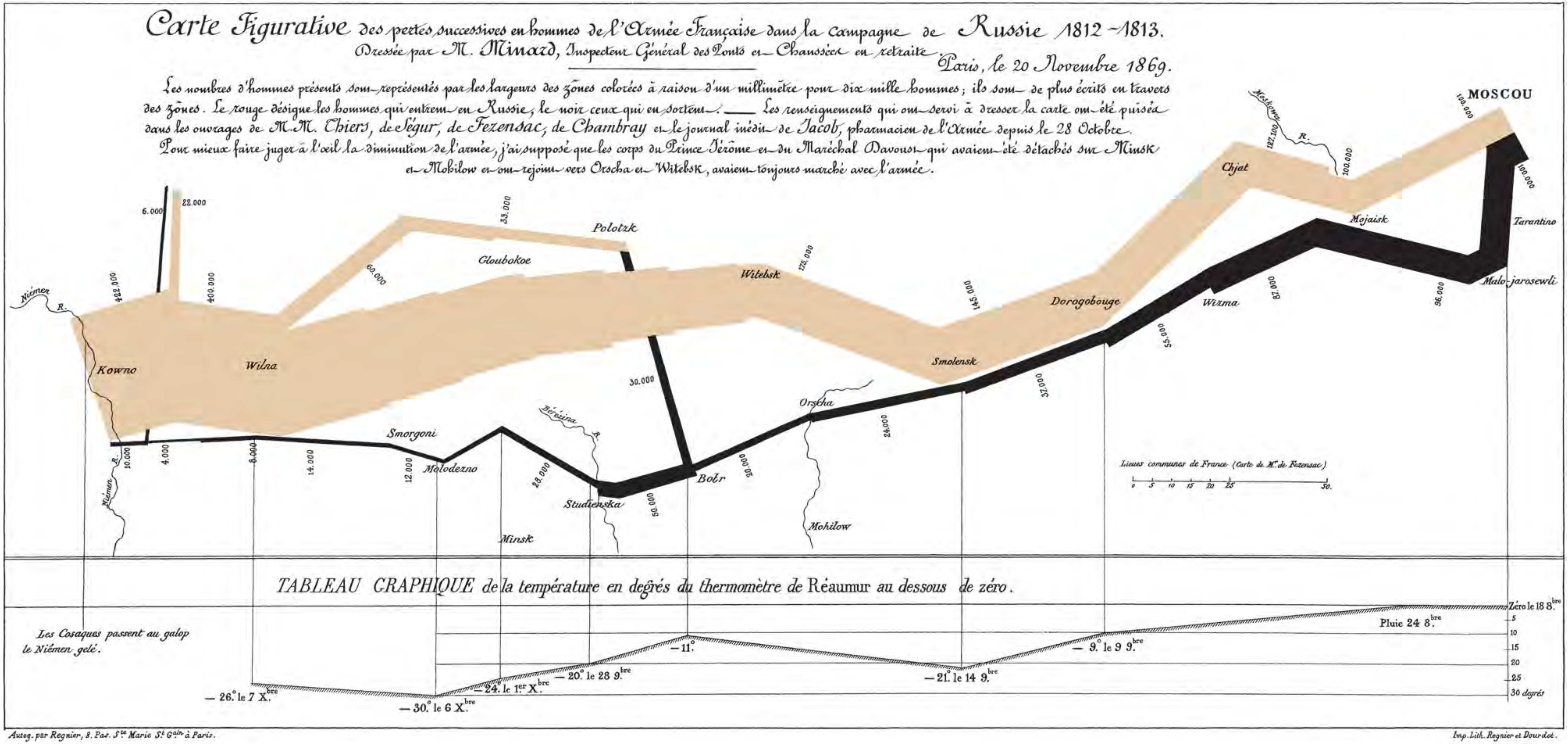
Policy reform and social justice through graphics





Quantitative data and geography; impact of the US Civil War on the cotton trade

Charles Minard; 1863



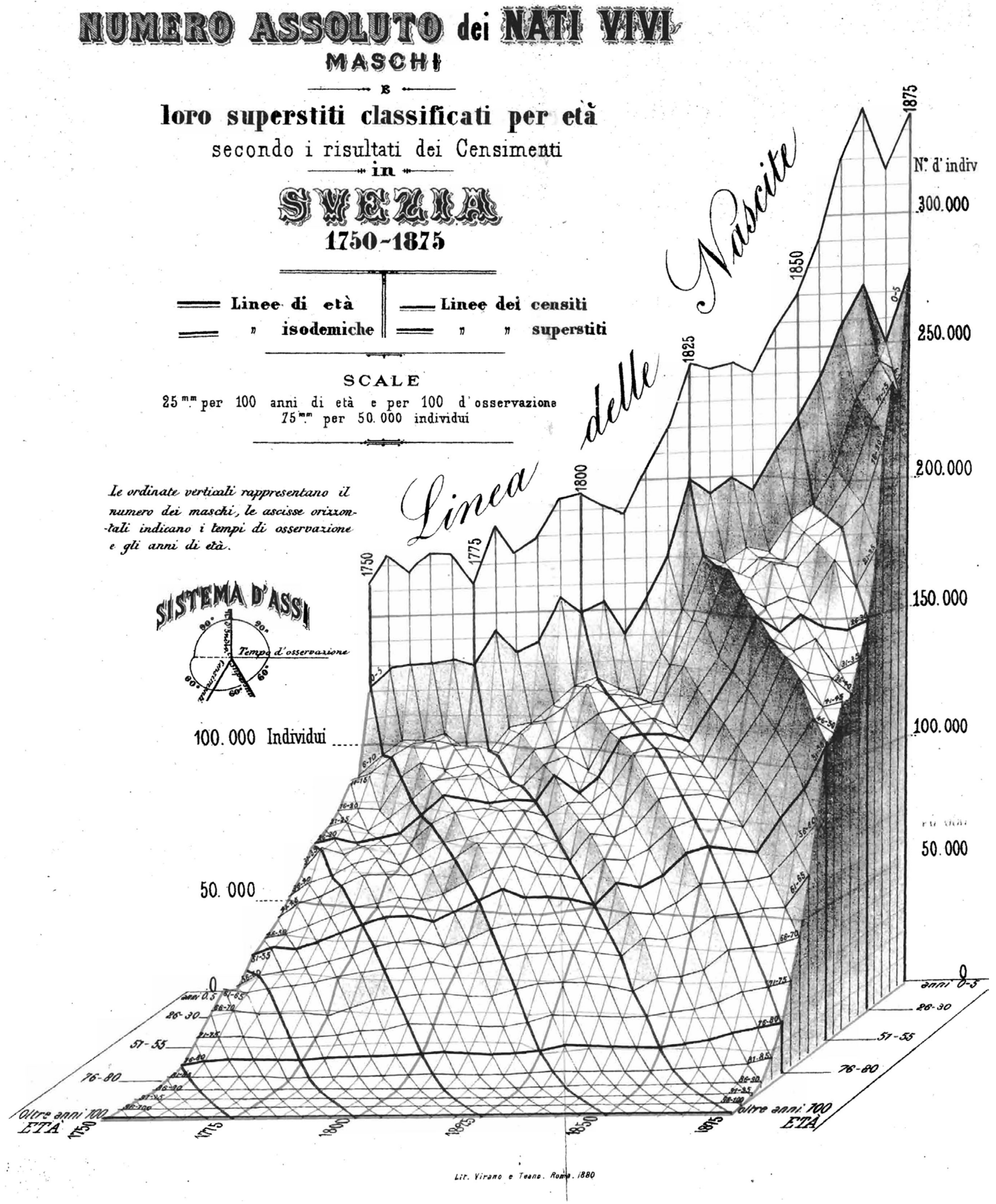
Telling a story with quantitative data: Napoleon’s march on Moscow; the first infographic?

Charles Minard; 1869

First topographic map;

contour lines to show third dimension

Marcellin du Carla-Boniface; 1782



Three variables in one plot;

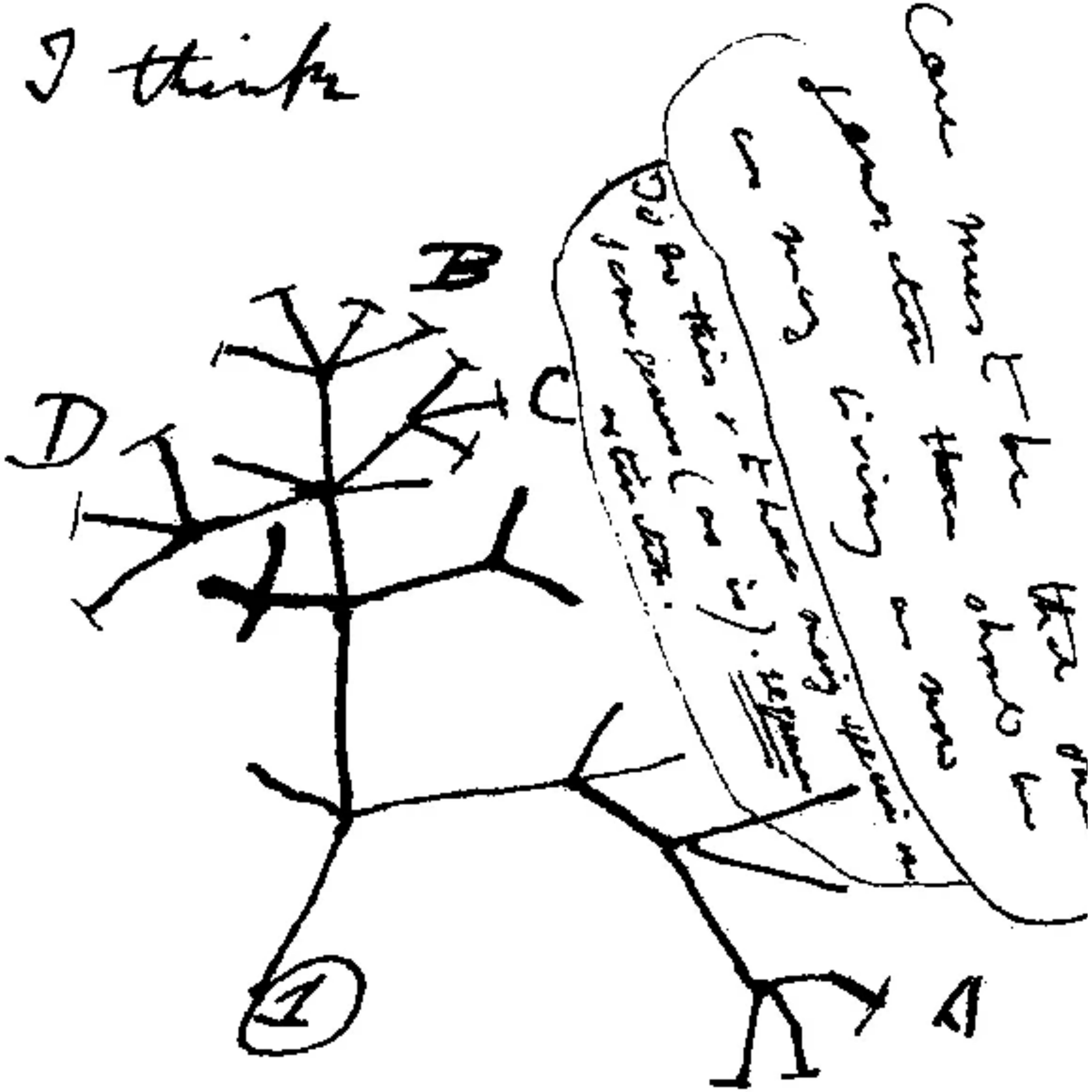
use of polygons and shading;

constructed by hand!

population of Sweden 1750-1875:

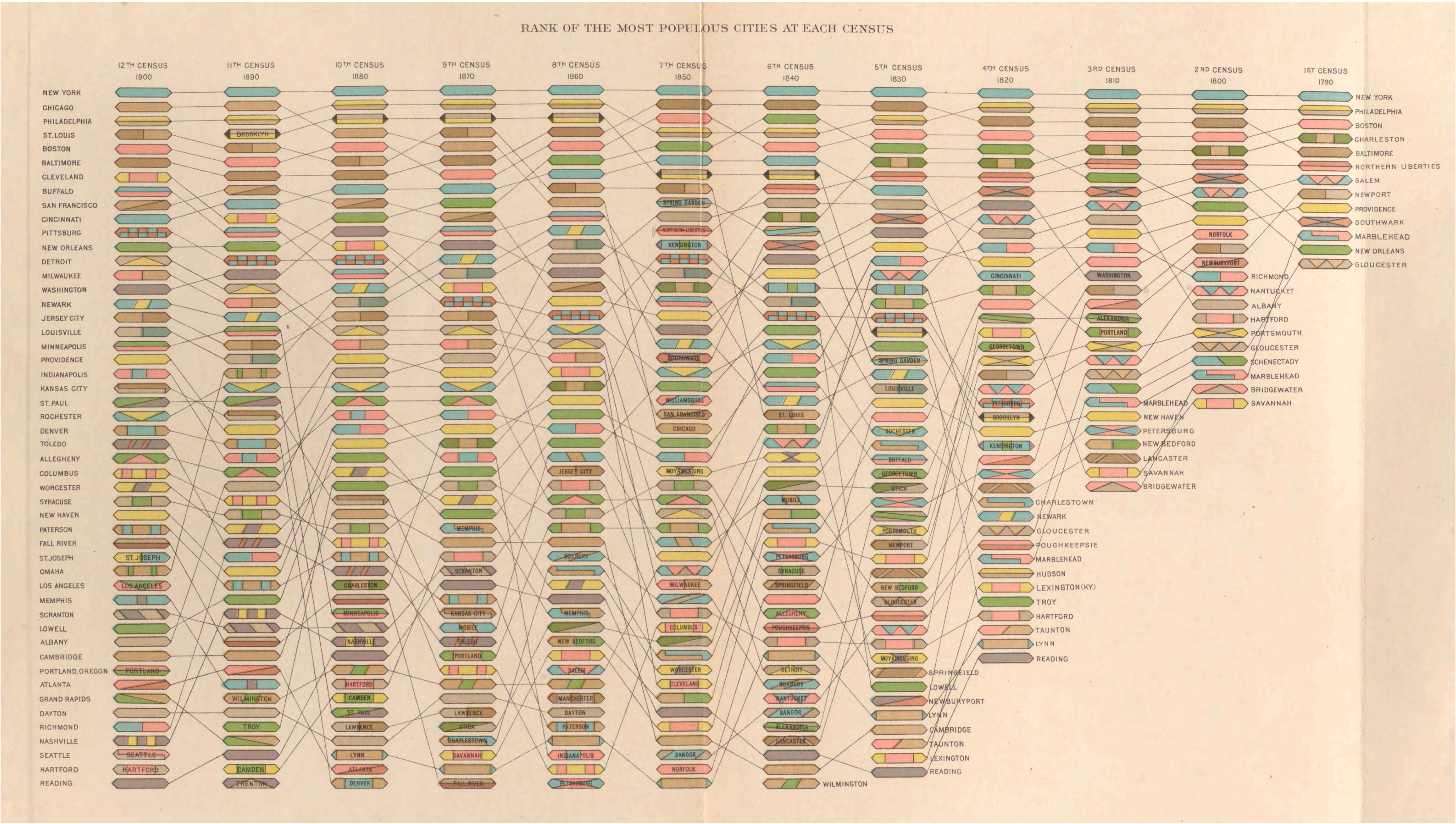
x = year, y = age, z = cumul. population

Luigi Perozzo; 1881



Diagramming relationships;
tree structure to indicate
genealogy

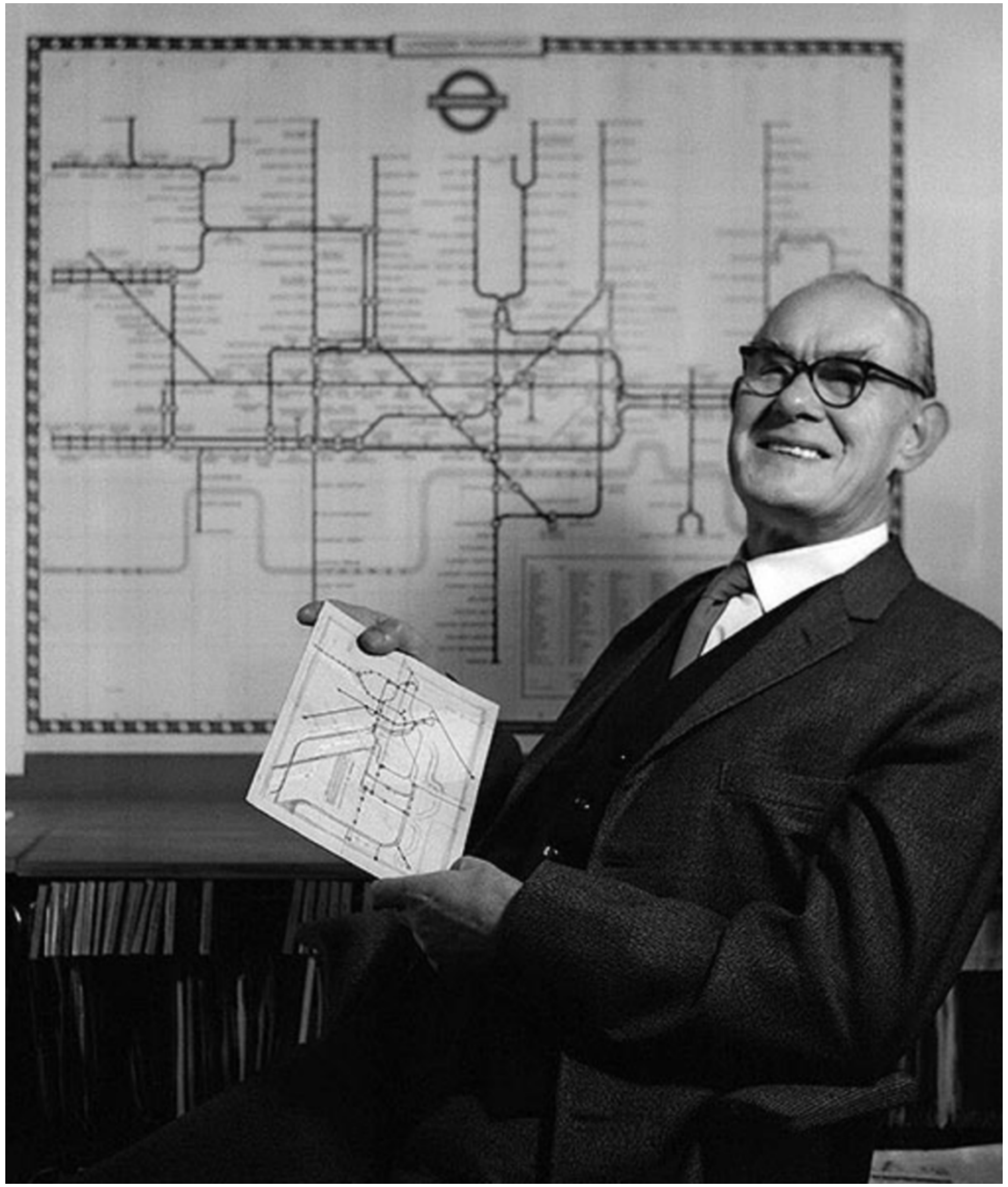
Charles Darwin; 1837



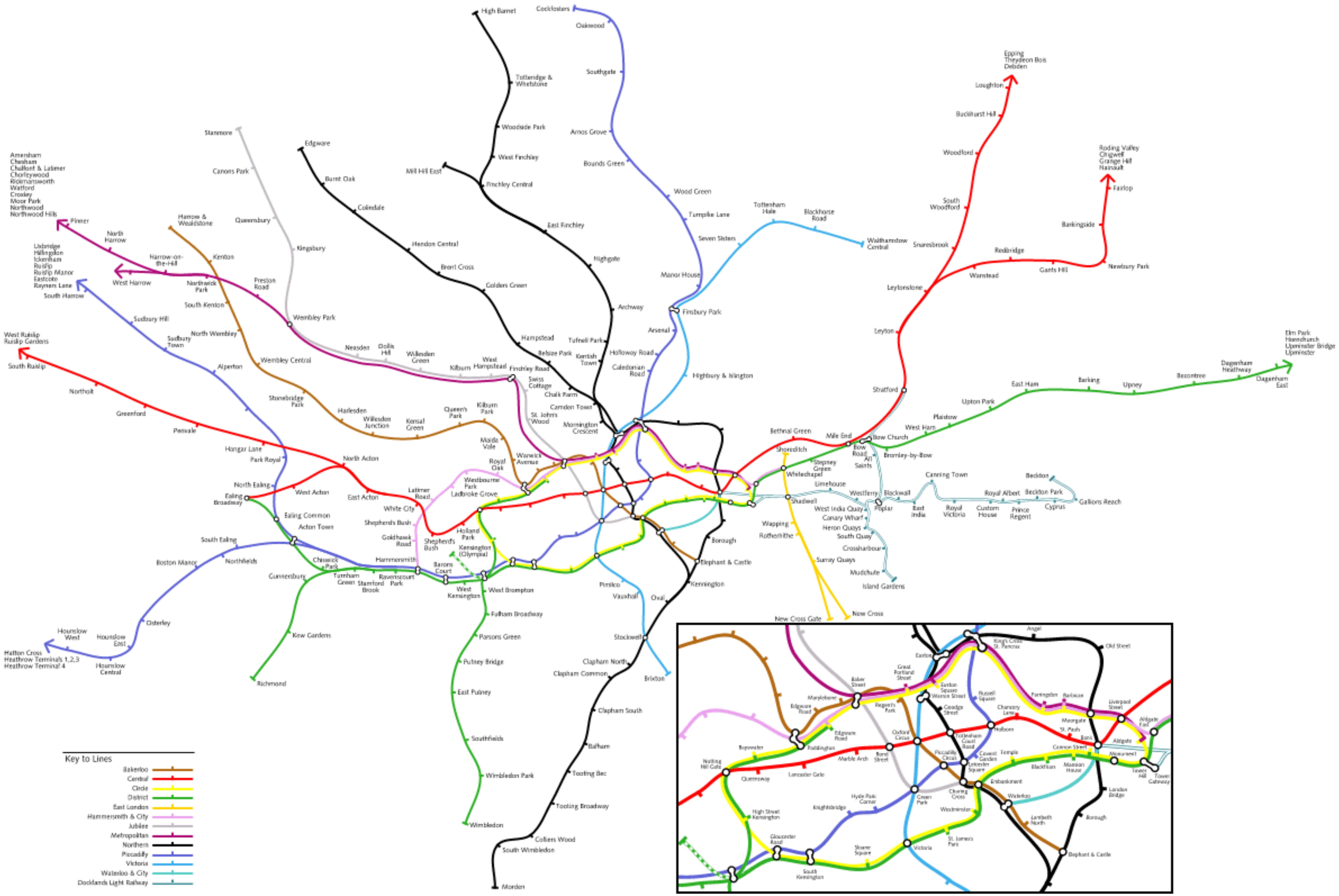
Visualizing changes in rank order; results of twelfth US census

Statistical Atlas; 1903

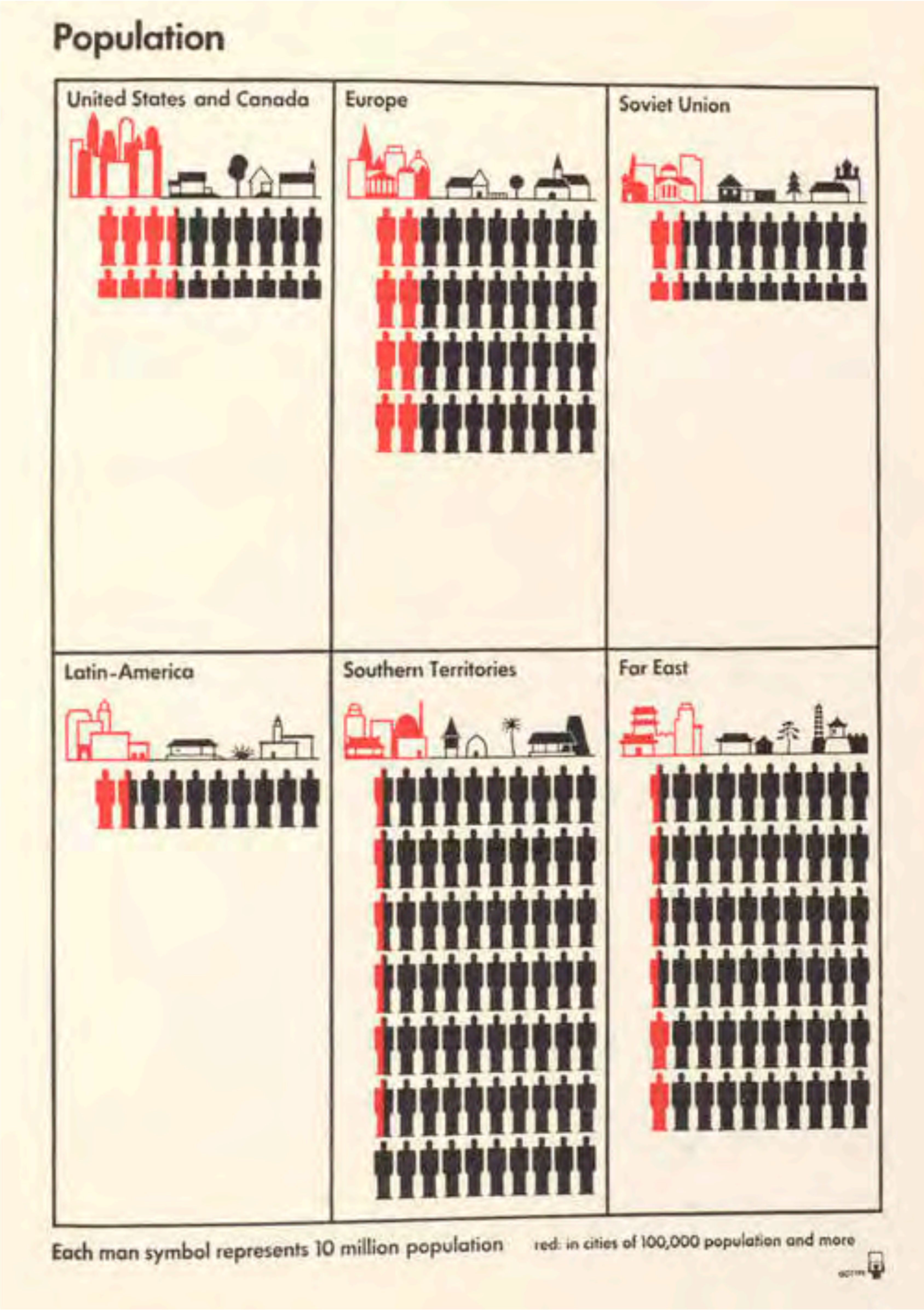
Practical, intelligent simplification

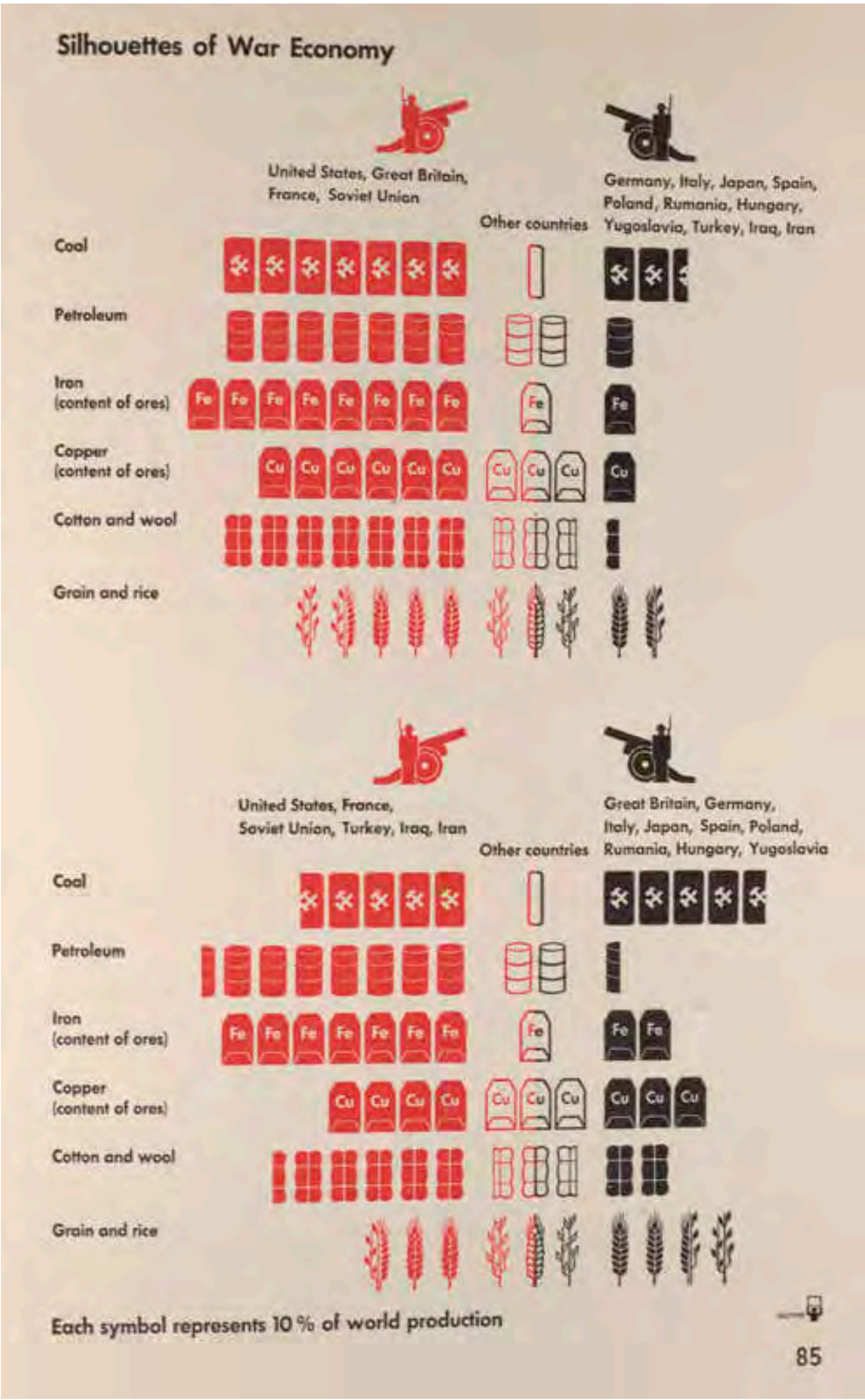


Henry “Harry” Beck; 1933



Geographically accurate London Underground map: not as easy to understand!



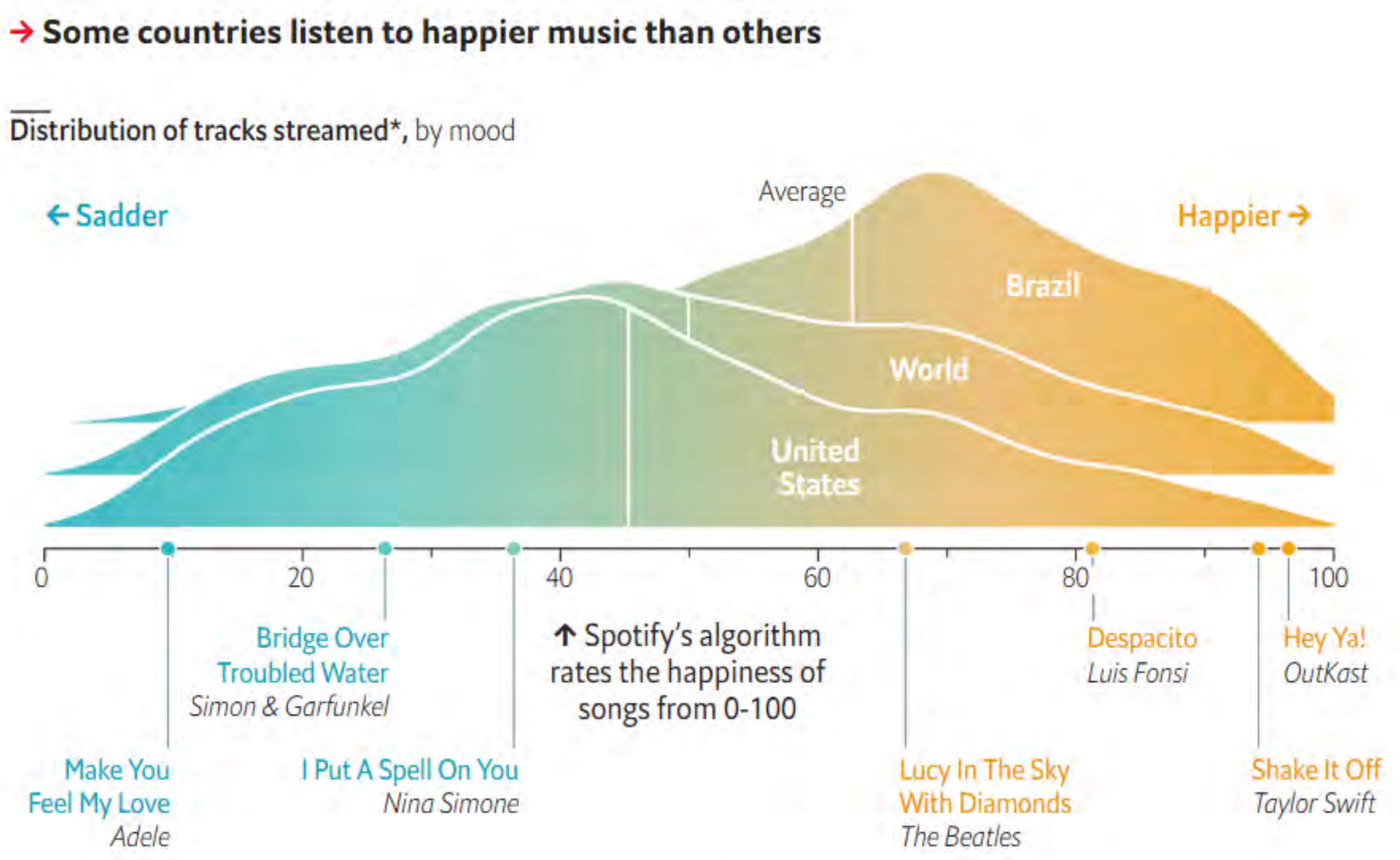


Symbols as an intuitive alternative to bar charts; uses color to group values

Otto Neurath; 1939

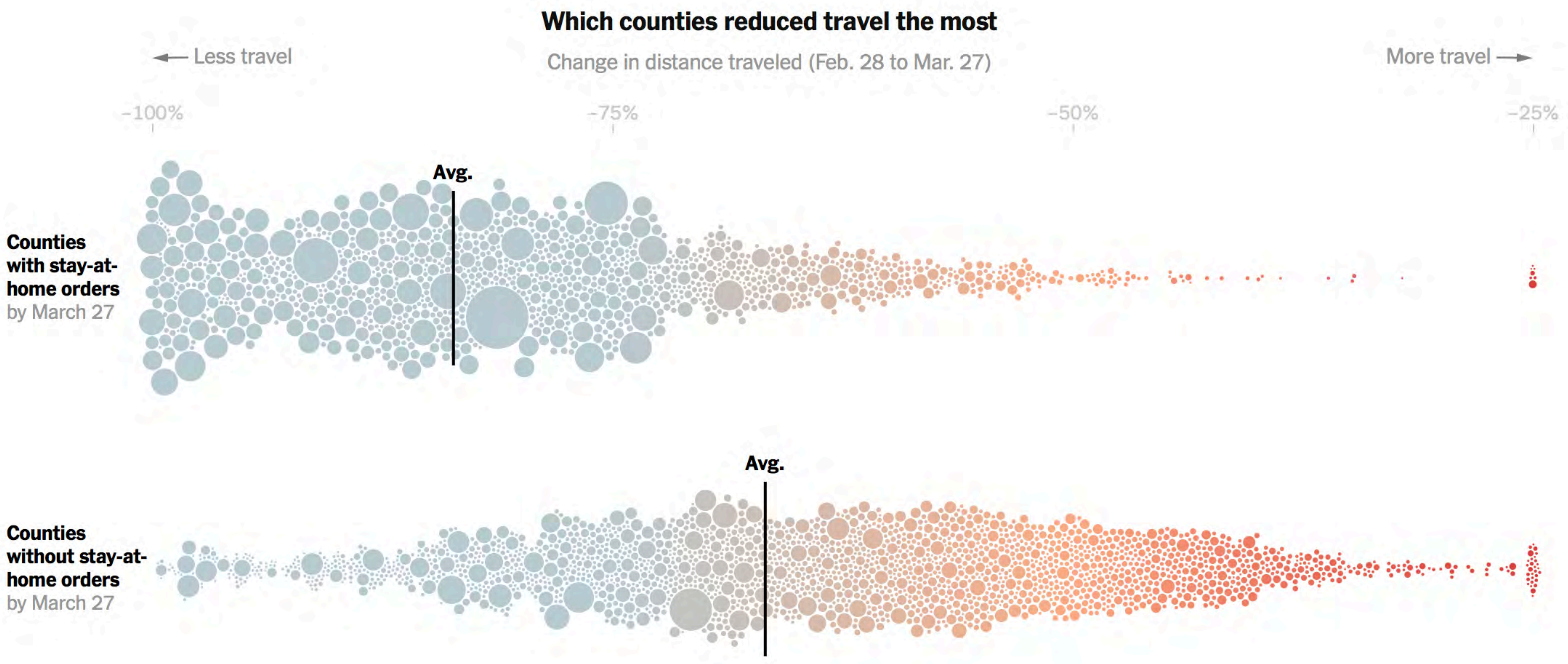
Modern Man in the Making

Director, International Foundation for Visual Education



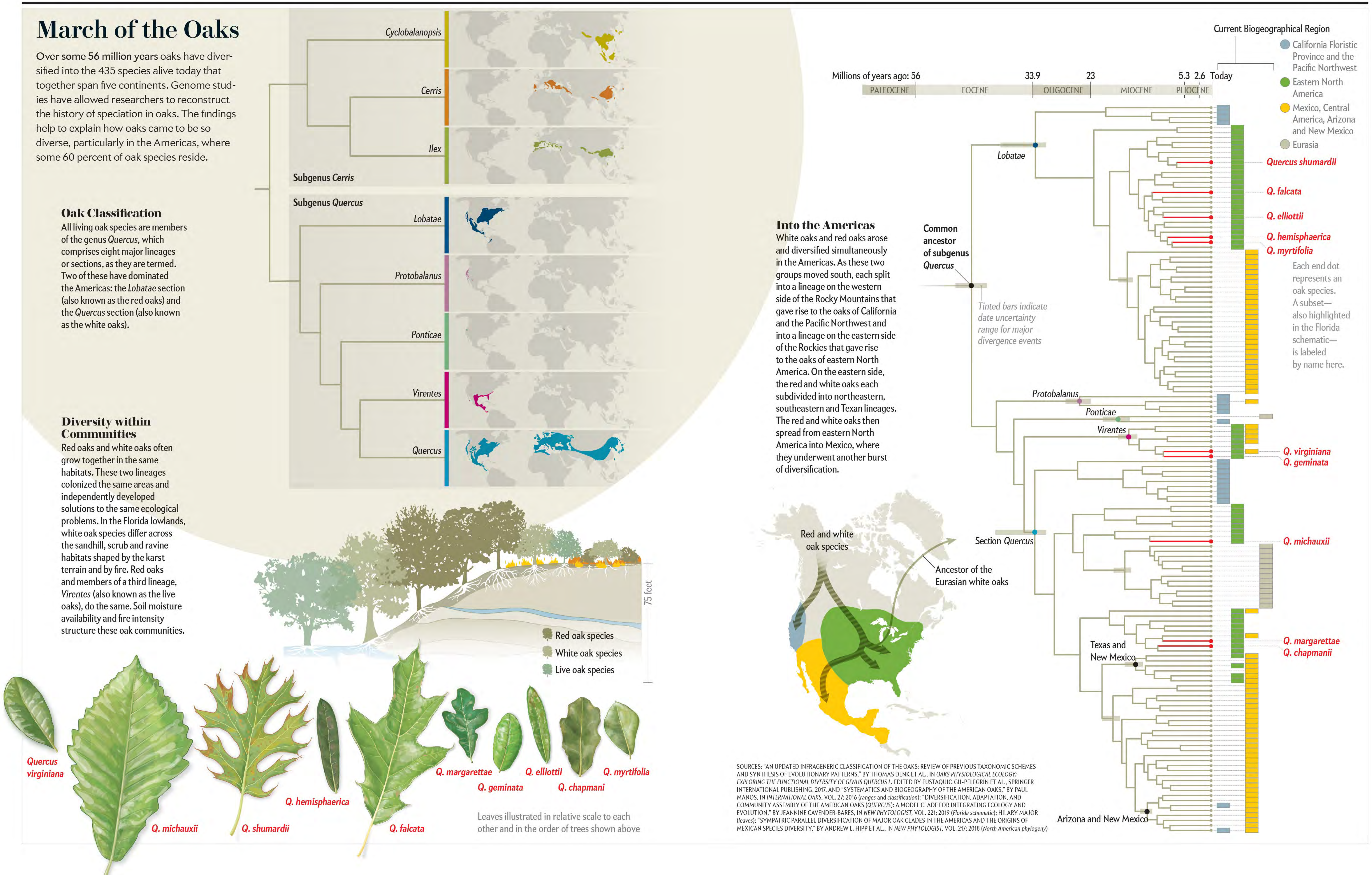
Ridge line plot; distribution of happiness score by country

Economist; 2020



Packed, jittered bubble plot: innovative ways of visualizing bivariate distributions

New York Times; 2020



March of the Oaks

Over some 56 million years oaks have diversified into the 435 species alive today that together span five continents. Genome studies have allowed researchers to reconstruct the history of speciation in oaks. The findings help to explain how oaks came to be so diverse, particularly in the Americas, where some 60 percent of oak species reside.

Oak Classification

All living oak species are members of the genus *Quercus*, which comprises eight major lineages or sections, as they are termed. Two of these have dominated the Americas: the *Lobatae* section (also known as the red oaks) and the *Quercus* section (also known as the white oaks).

Diversity within Communities

Red oaks and white oaks often grow together in the same habitats. These two lineages colonized the same areas and independently developed solutions to the same ecological problems. In the Florida lowlands, white oak species differ across the sandhill, scrub and ravine habitats shaped by the karst terrain and by fire. Red oaks and members of a third lineage, *Virentes* (also known as the live oaks), do the same. Soil moisture availability and fire intensity structure these oak communities.

Into the Americas

White oaks and red oaks arose and diversified simultaneously in the Americas. As these two groups moved south, each split into a lineage on the western side of the Rocky Mountains that gave rise to the oaks of California and the Pacific Northwest and into a lineage on the eastern side of the Rockies that gave rise to the oaks of eastern North America. On the eastern side, the red and white oaks each subdivided into northeastern, southeastern and Texan lineages. The red and white oaks then spread from eastern North America into Mexico, where they underwent another burst of diversification.

SOURCES: "AN UPDATED INFRAGENERIC CLASSIFICATION OF THE OAKS: REVIEW OF PREVIOUS TAXONOMIC SCHEMES AND SYNTHESIS OF EVOLUTIONARY PATTERNS," BY THOMAS DENK ET AL., IN *OAKS PHYSIOLOGICAL ECOLOGY: EXPLORING THE FUNCTIONAL DIVERSITY OF GENUS QUERCUS L.* EDITED BY EUSTAQUIO GIL-PELEGRÍN ET AL., SPRINGER INTERNATIONAL PUBLISHING, 2017, AND "SYSTEMATICS AND BIOGEOGRAPHY OF THE AMERICAN OAKS," BY PAUL MANOS, IN *INTERNATIONAL OAKS,* VOL. 27; 2016 (*ranges* and *classification*); "DIVERSIFICATION, ADAPTATION, AND COMMUNITY ASSEMBLY OF THE AMERICAN OAKS (*QUERCUS*): A MODEL CLADE FOR INTEGRATING ECOLOGY AND EVOLUTION," BY JEANNINE CAVENDER-BARES, IN *NEW PHYTOLOGIST,* VOL. 221; 2019 (*Florida schematic*); HILARY MAJOR (*leaves*); "SYMPATRIC PARALLEL DIVERSIFICATION OF MAJOR OAK CLADES IN THE AMERICAS AND THE ORIGINS OF MEXICAN SPECIES DIVERSITY," BY ANDREW L. HIPP ET AL., IN *NEW PHYTOLOGIST,* VOL. 217; 2018 (*North American phylogeny*)

Infographic of oak evolution   Paul Manos (Duke Biology); 2020

Sources

History of Data Visualization: https://friendly.github.io/HistDataVis/

Milestones in the History of … Data Visualization: https://www.datavis.ca/milestones/index.php?page=home

Wikipedia: https://en.wikipedia.org/wiki/Data_and_information_visualization

National Library of Medicine, NIH: https://circulatingnow.nlm.nih.gov/2023/07/13/revealing-data-visualizations-in-historical-collections/

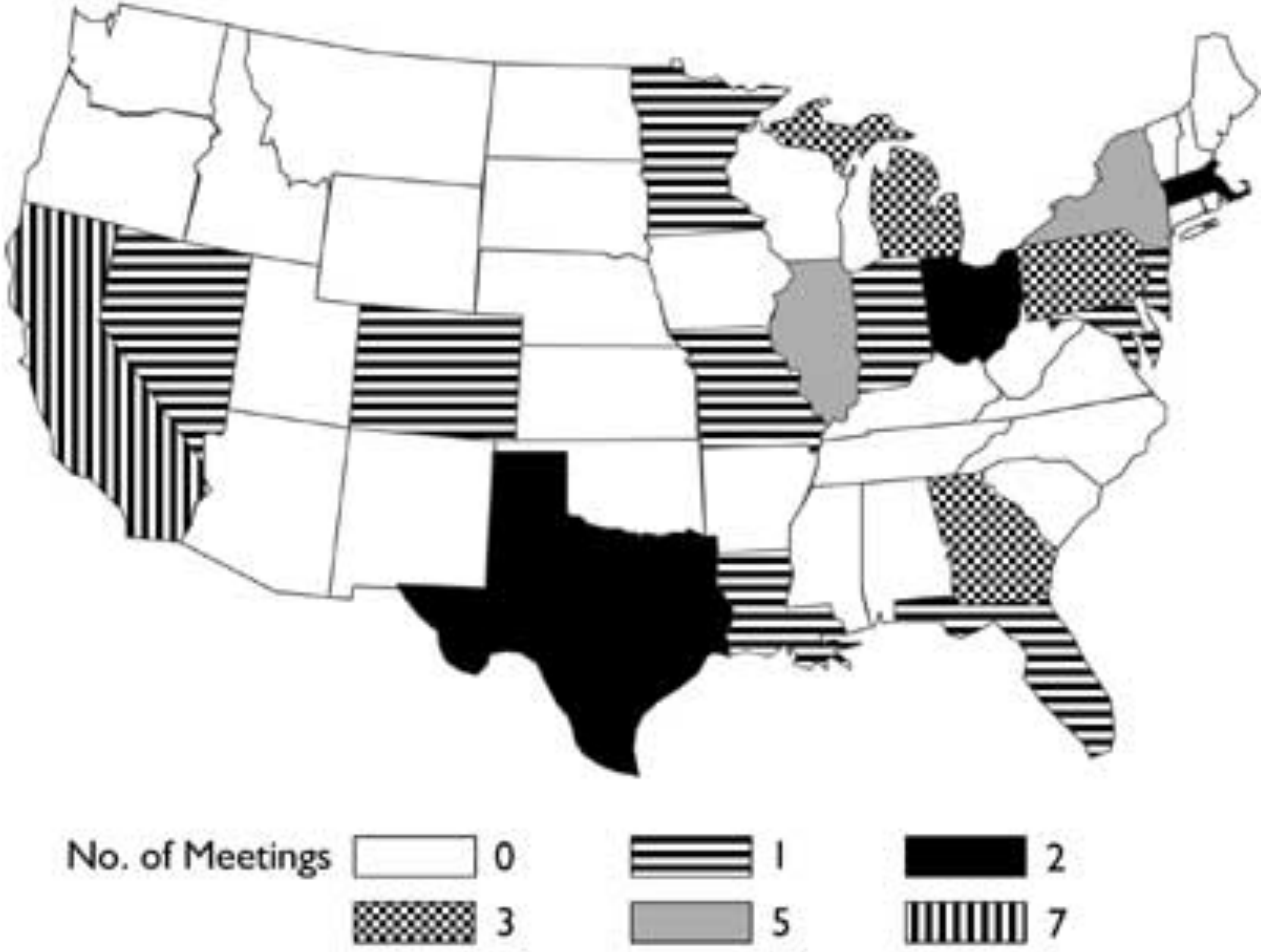
panoply: https://blog.panoply.io/history-of-data-visualization

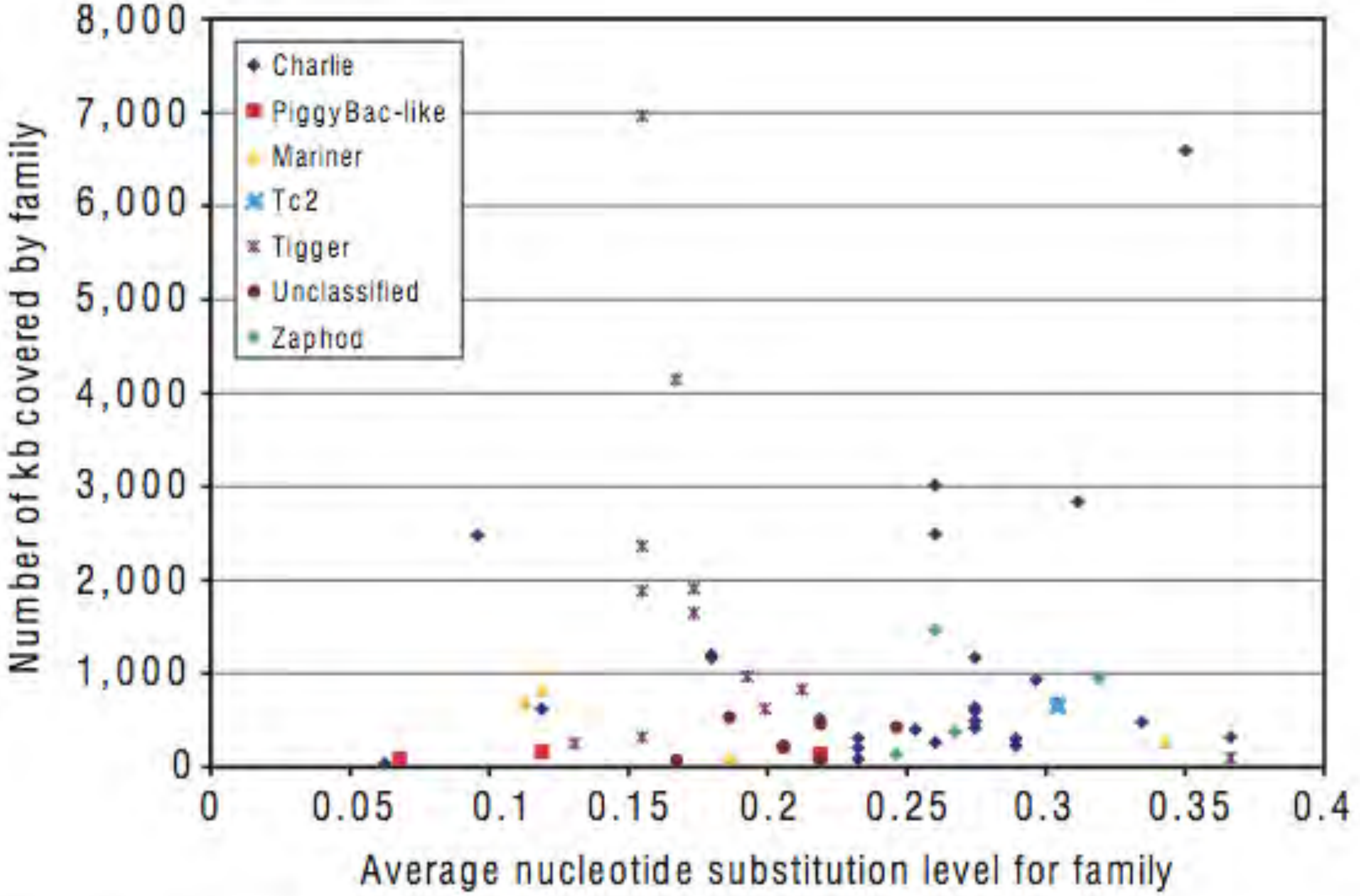
The Data School: https://www.thedataschool.co.uk/luke-bennett/a-history-of-data-visualization-part-1-ancient-analysts/

A Brief History of Visualization: https://link.springer.com/chapter/10.1007/978-3-540-33037-0_2

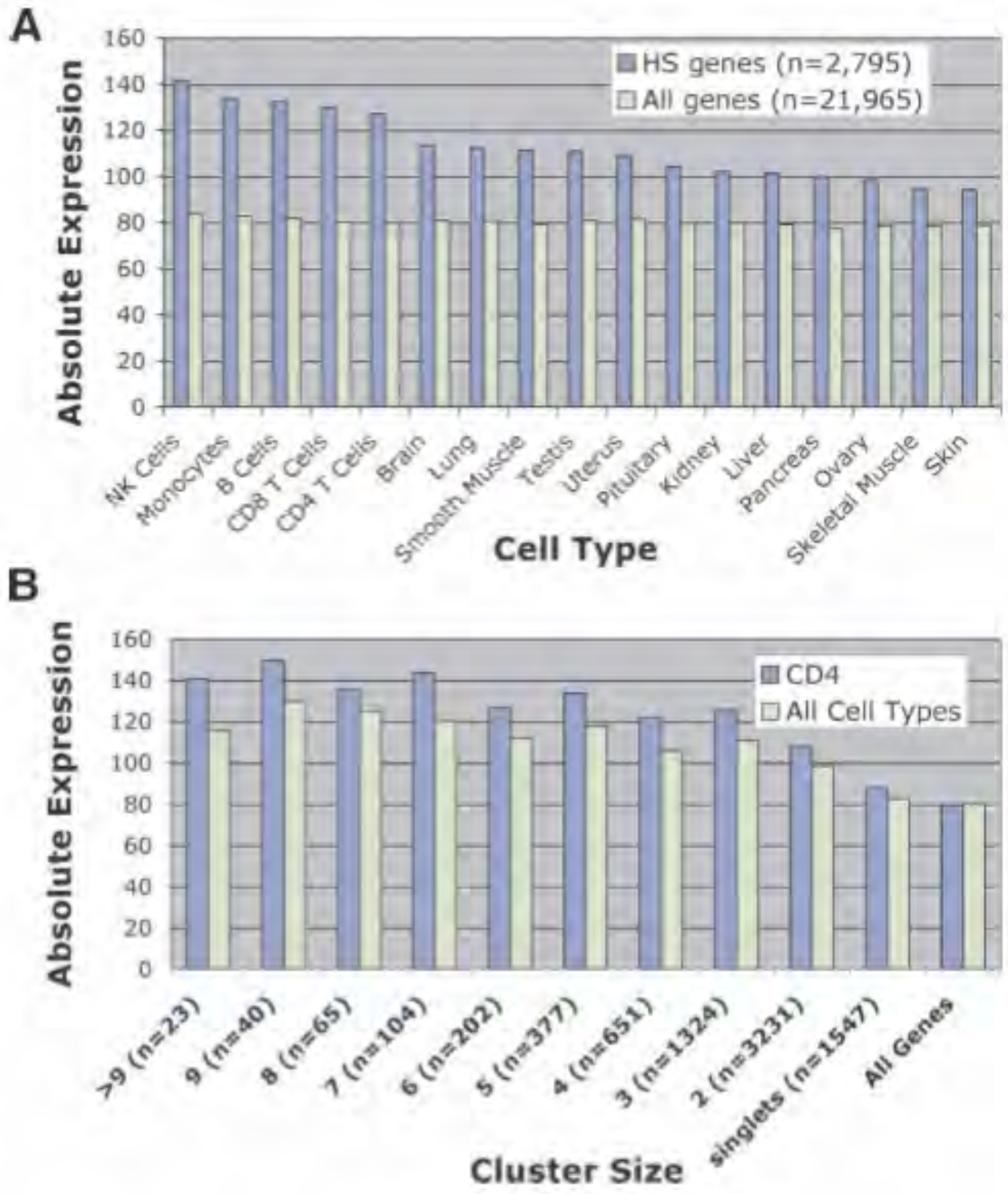
Tableau: https://www.tableau.com/visualization/data-visualization-examples

Counter-examples of graphical excellence

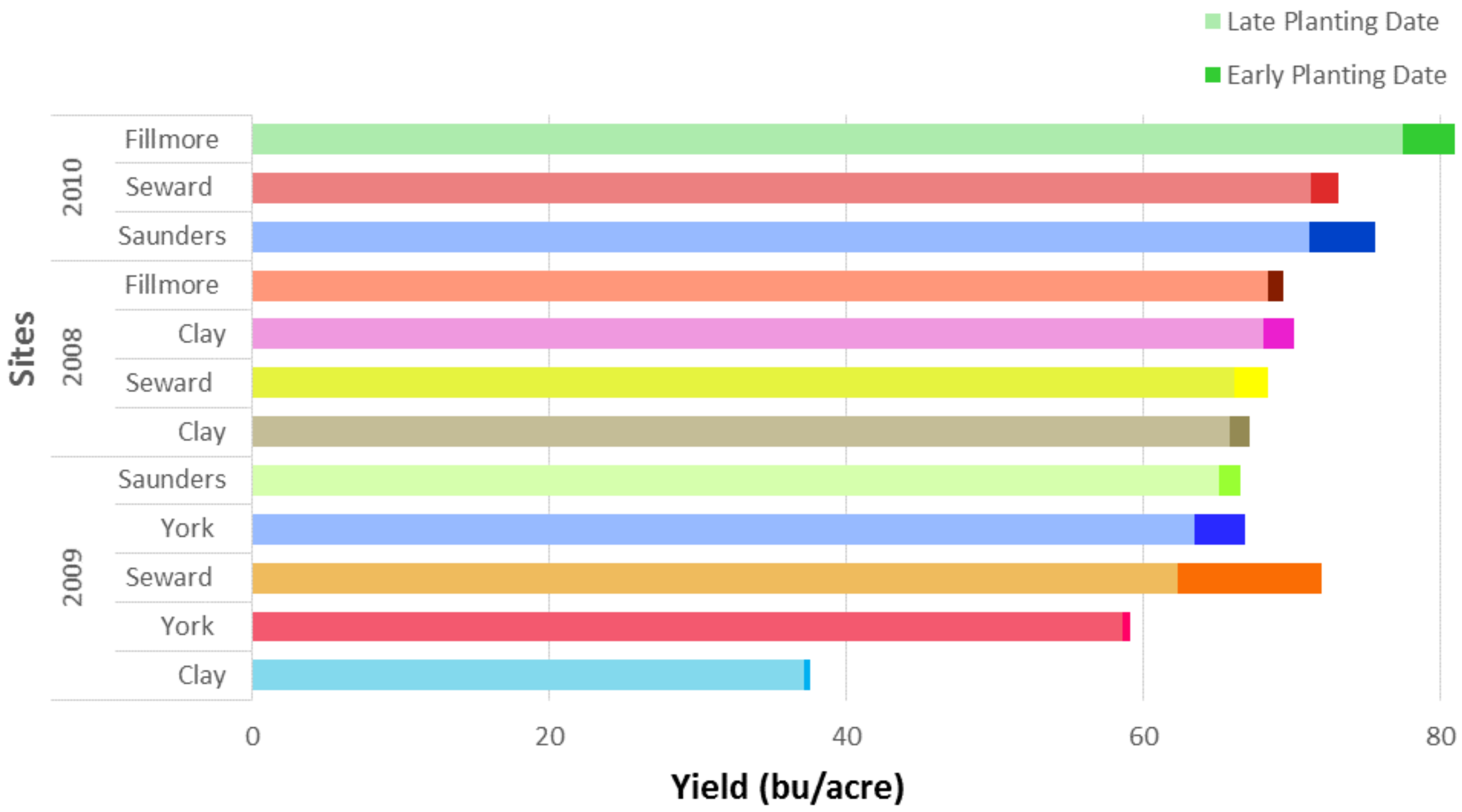

No. of Meetings
0
1
2
3
5
7



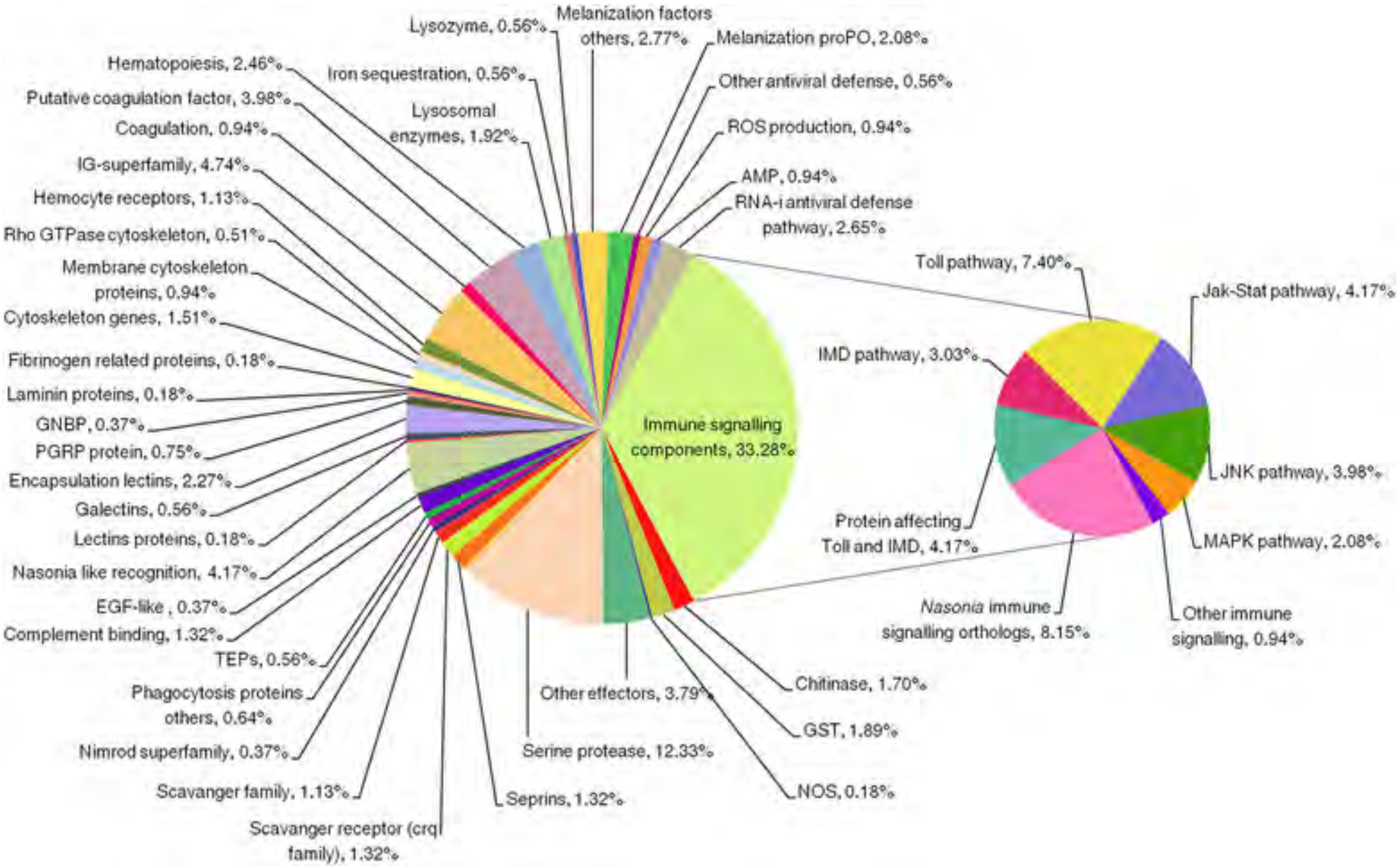
8,000
7,000
6,000
5,000
4,000
3,000
2,000
1,000
0
Number of kb covered by family
Charlie
PiggyBac-like
Mariner
Tc2
Tigger
Unclassified
Zaphod
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
Average nucleotide substitution level for family

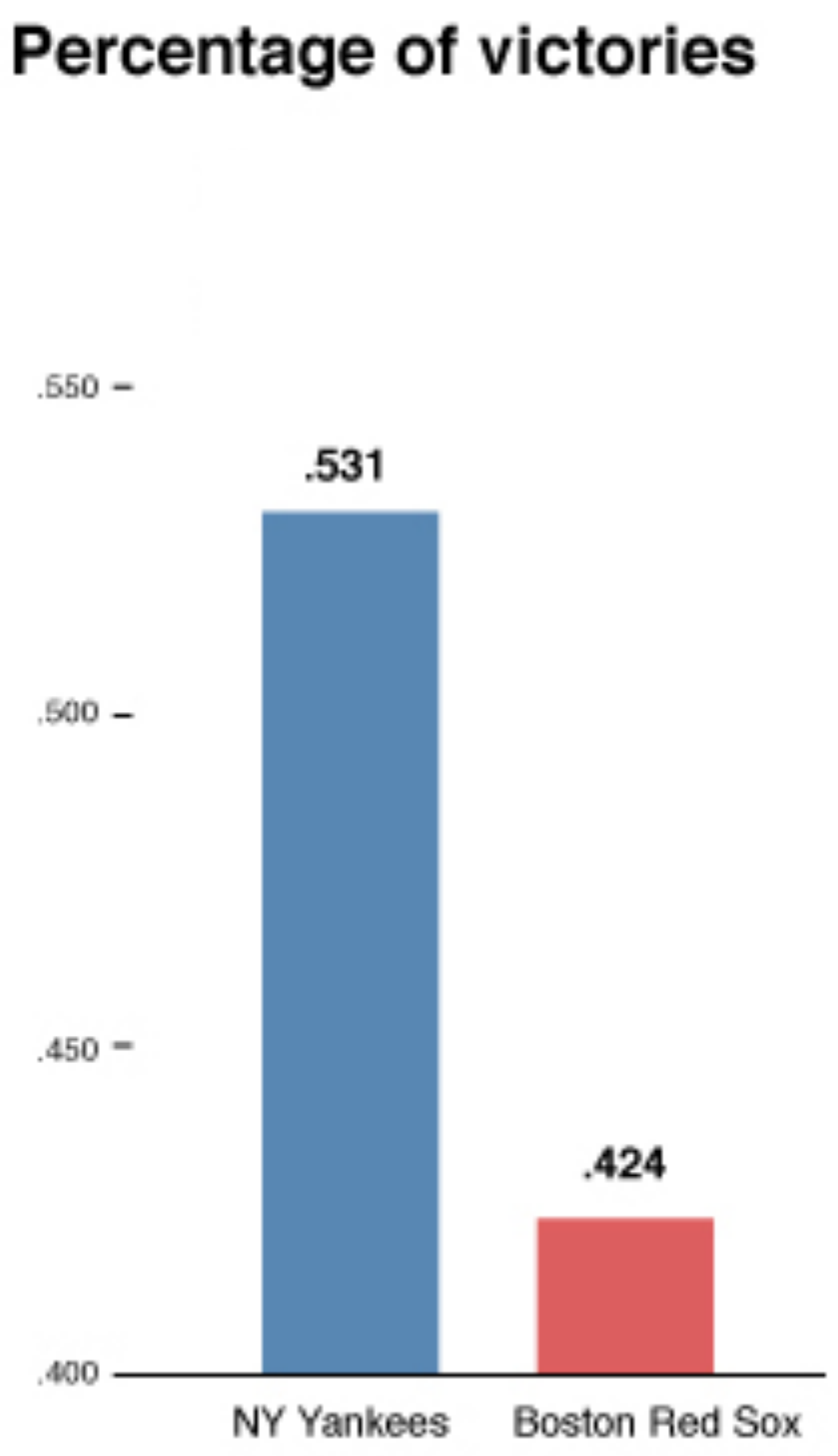


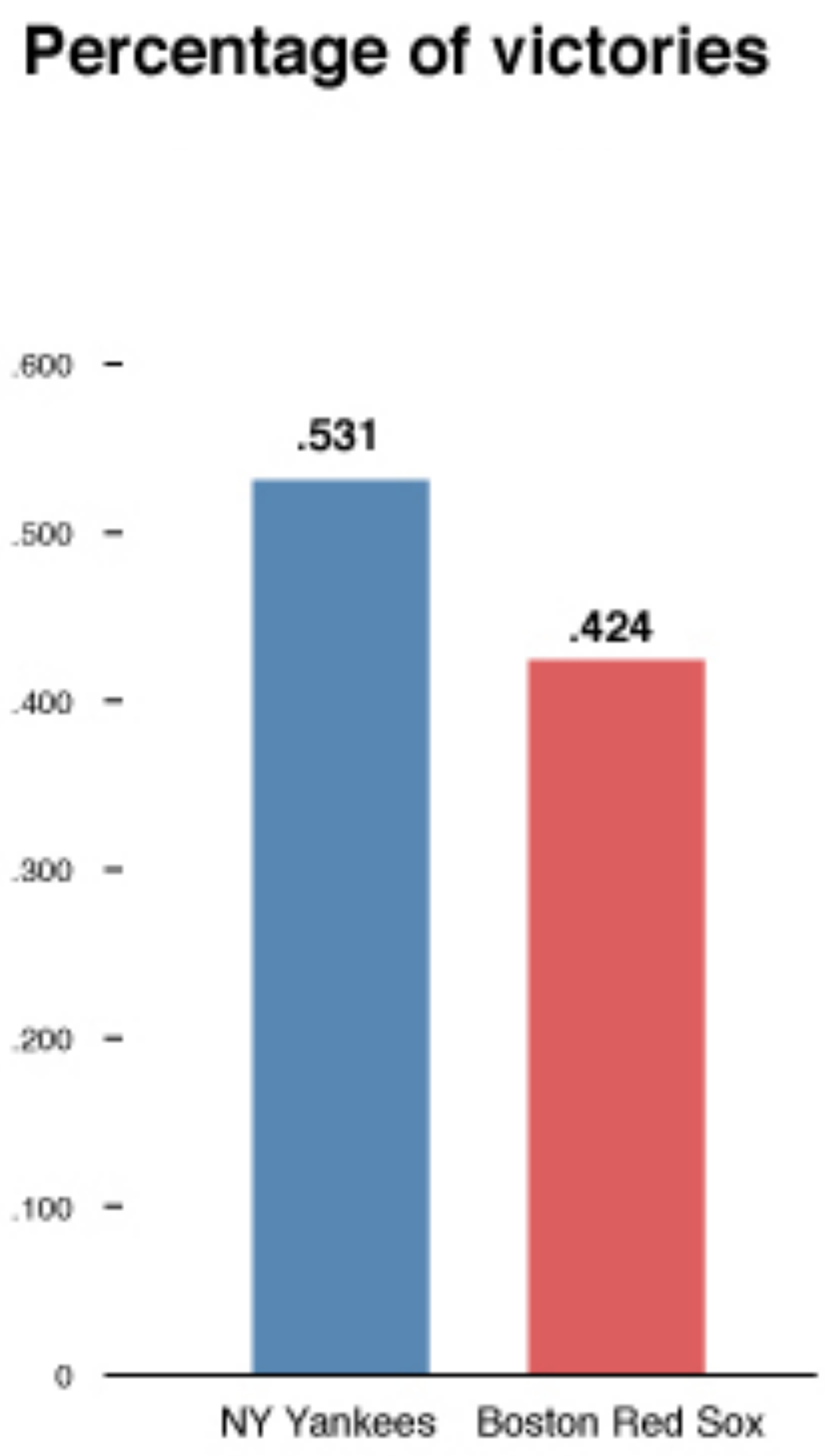
A
Absolute Expression
HS genes (n=2,795)
All genes (n=21,965)
Cell Type
NK Cells
Monocytes
B Cells
CD8 T Cells
CD4 T Cells
Brain
Lung
Smooth Muscle
Testis
Uterus
Pituitary
Kidney
Liver
Pancreas
Ovary
Skeletal Muscle
Skin
B
Absolute Expression
CD4
All Cell Types
Cluster Size
>9 (n=23)
9 (n=40)
8 (n=65)
7 (n=104)
6 (n=202)
5 (n=377)
4 (n=651)
3 (n=1324)
2 (n=3231)
singlets (n=1547)
All Genes

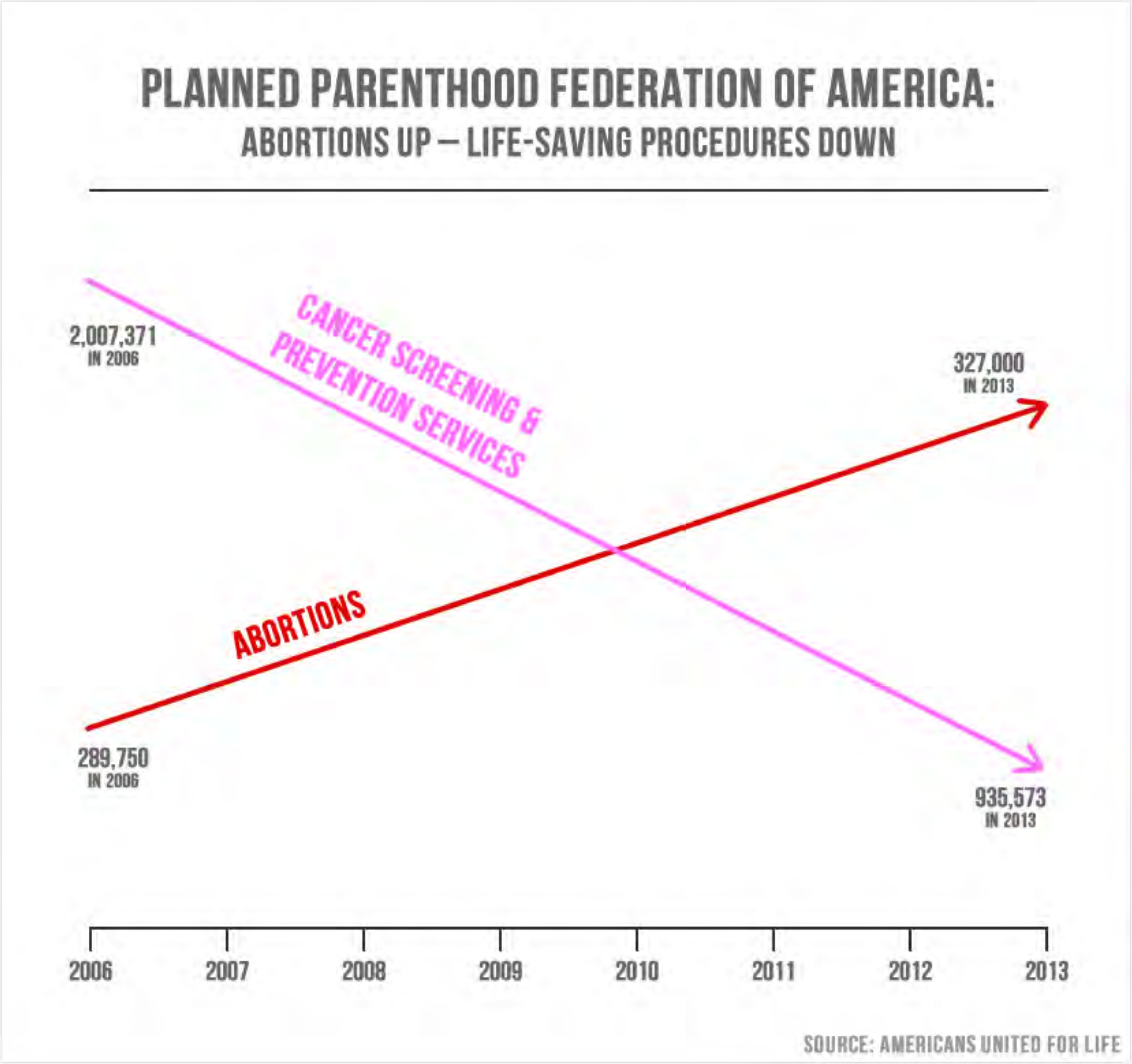


Late Planting Date
Early Planting Date
Sites
2010
Fillmore
Seward
Saunders
2008
Fillmore
Clay
Seward
Clay
2009
Saunders
York
Seward
York
Clay
0
20
40
60
80
Yield (bu/acre)

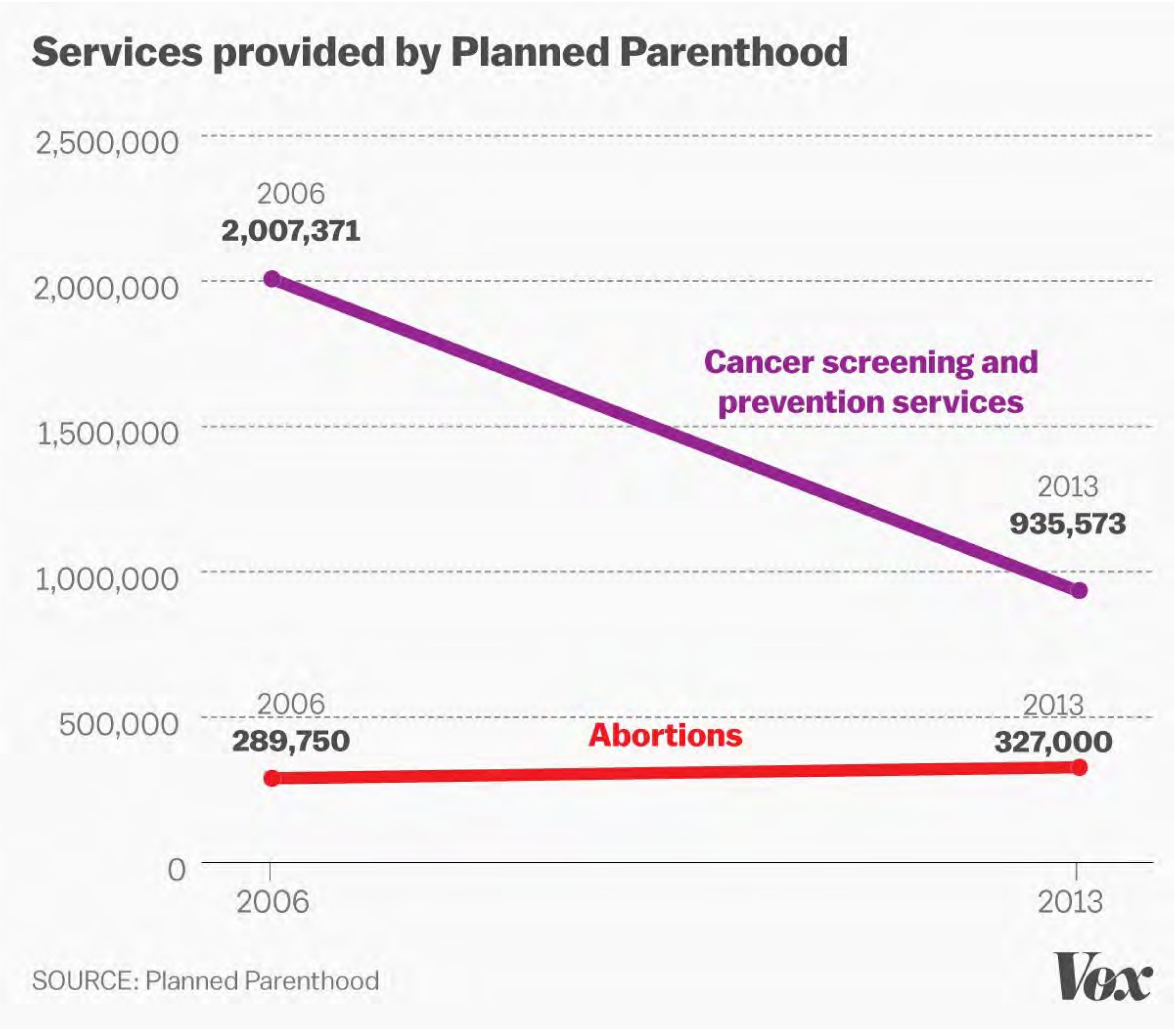

Lysozyme, 0.56%
Melanization factors others, 2.77%
Melanization proPO, 2.08%
Hematopoiesis, 2.46%
Iron sequestration, 0.56%
Other antiviral defense, 0.56%
Putative coagulation factor, 3.98%
Lysosomal enzymes, 1.92%
Coagulation, 0.94%
ROS production, 0.94%
IG-superfamily, 4.74%
AMP, 0.94%
Hemocyte receptors, 1.13%
RNA-i antiviral defense pathway, 2.65%
Rho GTPase cytoskeleton, 0.51%
Membrane cytoskeleton proteins, 0.94%
Toll pathway, 7.40%
Jak-Stat pathway, 4.17%
Cytoskeleton genes, 1.51%
IMD pathway, 3.03%
Fibrinogen related proteins, 0.18%
Laminin proteins, 0.18%
GNBP, 0.37%
Immune signalling components, 33.28%
PGRP protein, 0.75%
JNK pathway, 3.98%
Encapsulation lectins, 2.27%
Galectins, 0.56%
Protein affecting Toll and IMD, 4.17%
Lectins proteins, 0.18%
MAPK pathway, 2.08%
Nasonia like recognition, 4.17%
EGF-like, 0.37%
Nasonia immune signalling orthologs, 8.15%
Other immune signalling, 0.94%
Complement binding, 1.32%
TEPs, 0.56%
Chitinase, 1.70%
Phagocytosis proteins others, 0.64%
Other effectors, 3.79%
GST, 1.89%
Nimrod superfamily, 0.37%
Serine protease, 12.33%
Scavanger family, 1.13%
NOS, 0.18%
Seprins, 1.32%
Scavanger receptor (crq family), 1.32%


Percentage of victories
.550
.500
.450
.400
.531
.424
NY Yankees
Boston Red Sox

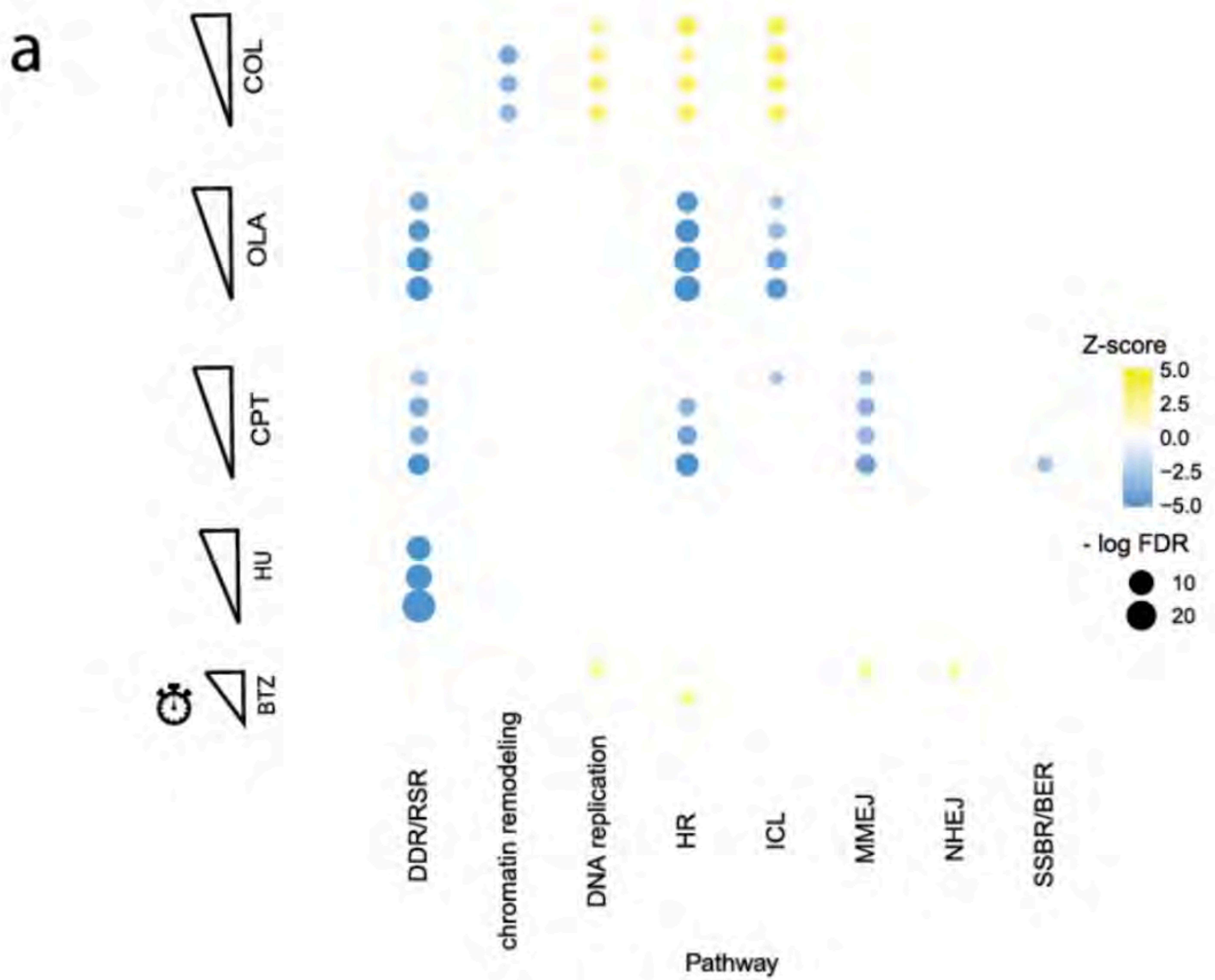

Percentage of victories
.600
.500
.400
.300
.200
.100
0
.531
.424
NY Yankees
Boston Red Sox

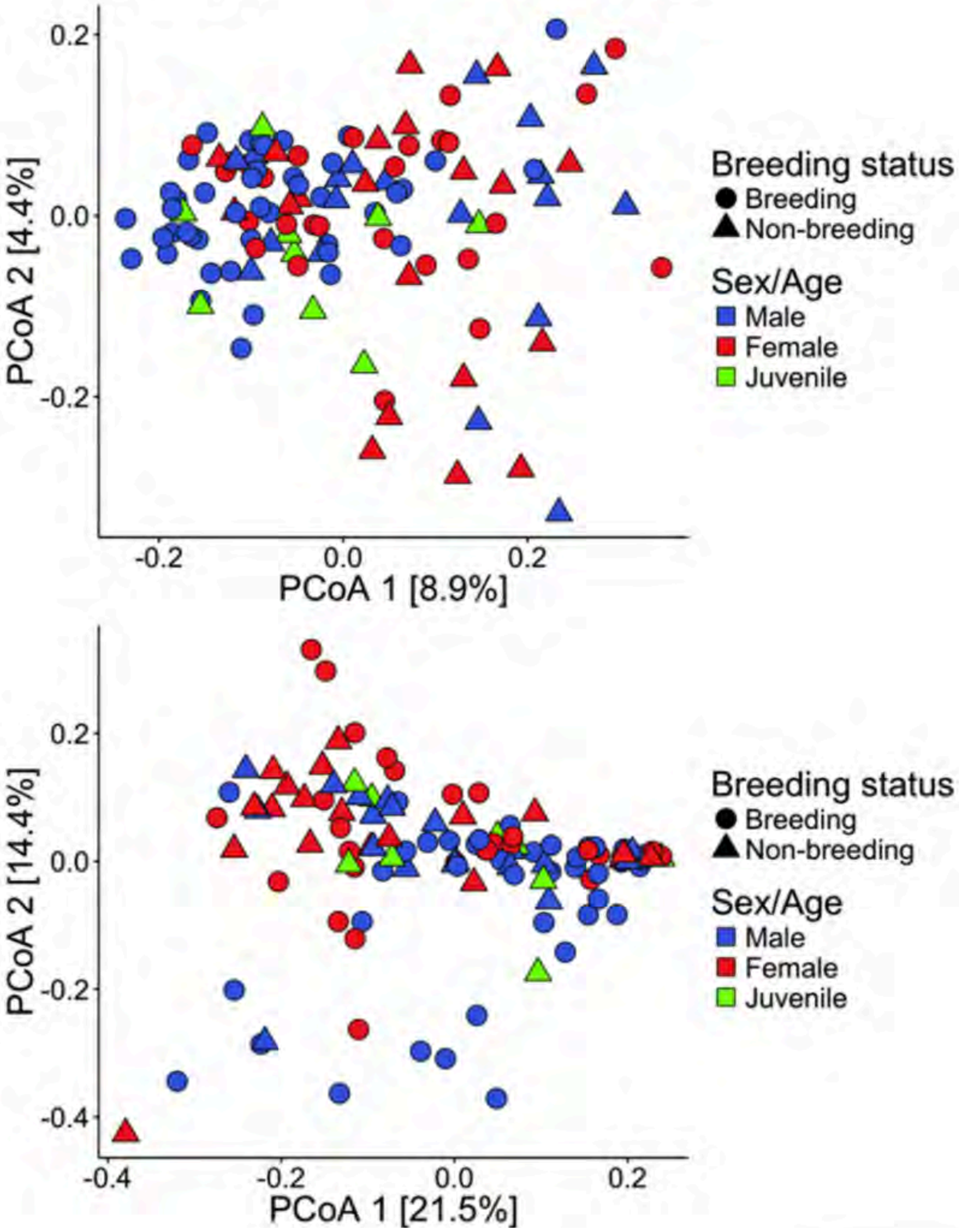


PLANNED PARENTHOOD FEDERATION OF AMERICA:
ABORTIONS UP – LIFE-SAVING PROCEDURES DOWN
2,007,371
IN 2006
CANCER SCREENING &
PREVENTION SERVICES
327,000
IN 2013
ABORTIONS
289,750
IN 2006
935,573
IN 2013
2006
2007
2008
2009
2010
2011
2012
2013
SOURCE: AMERICANS UNITED FOR LIFE

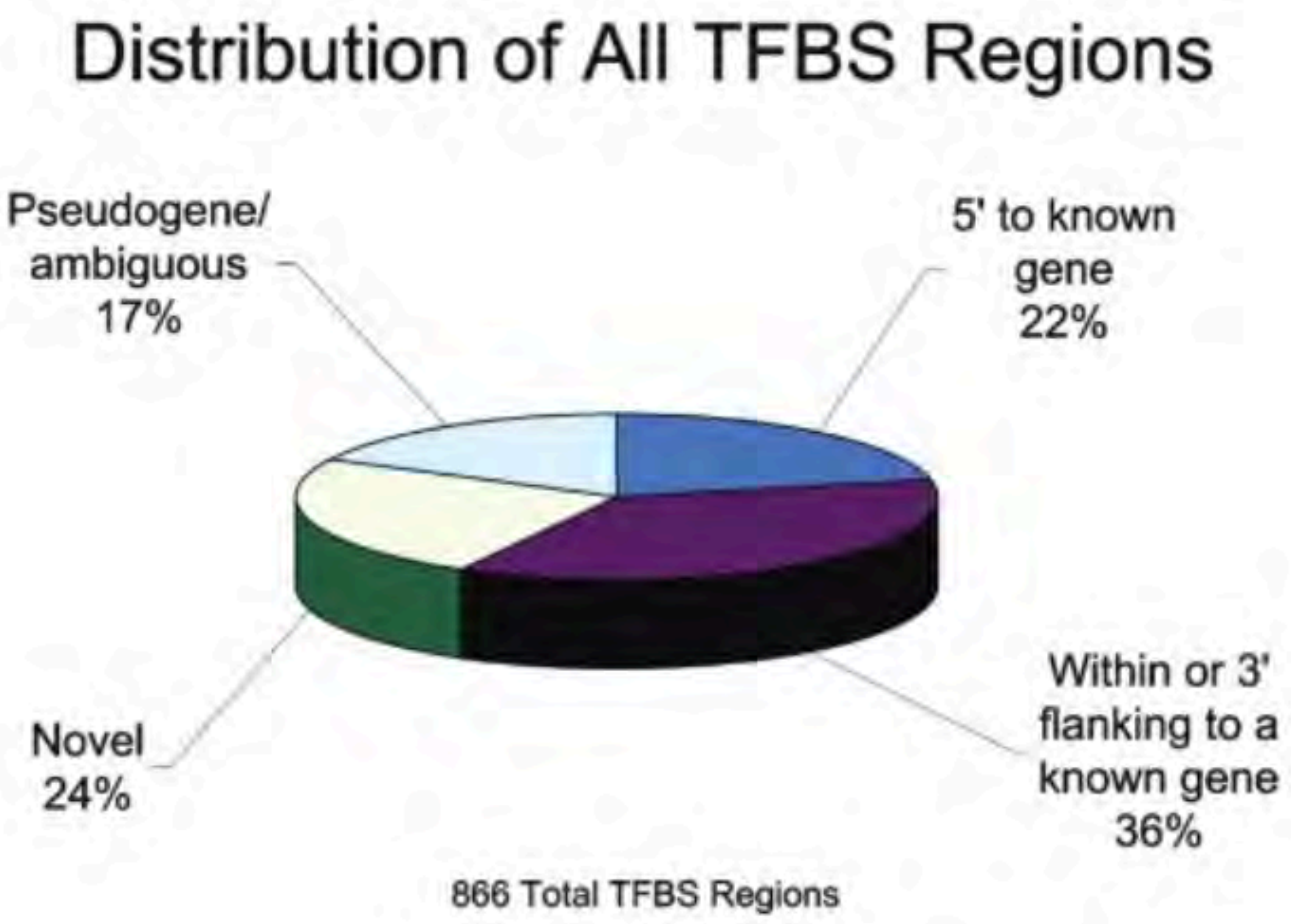


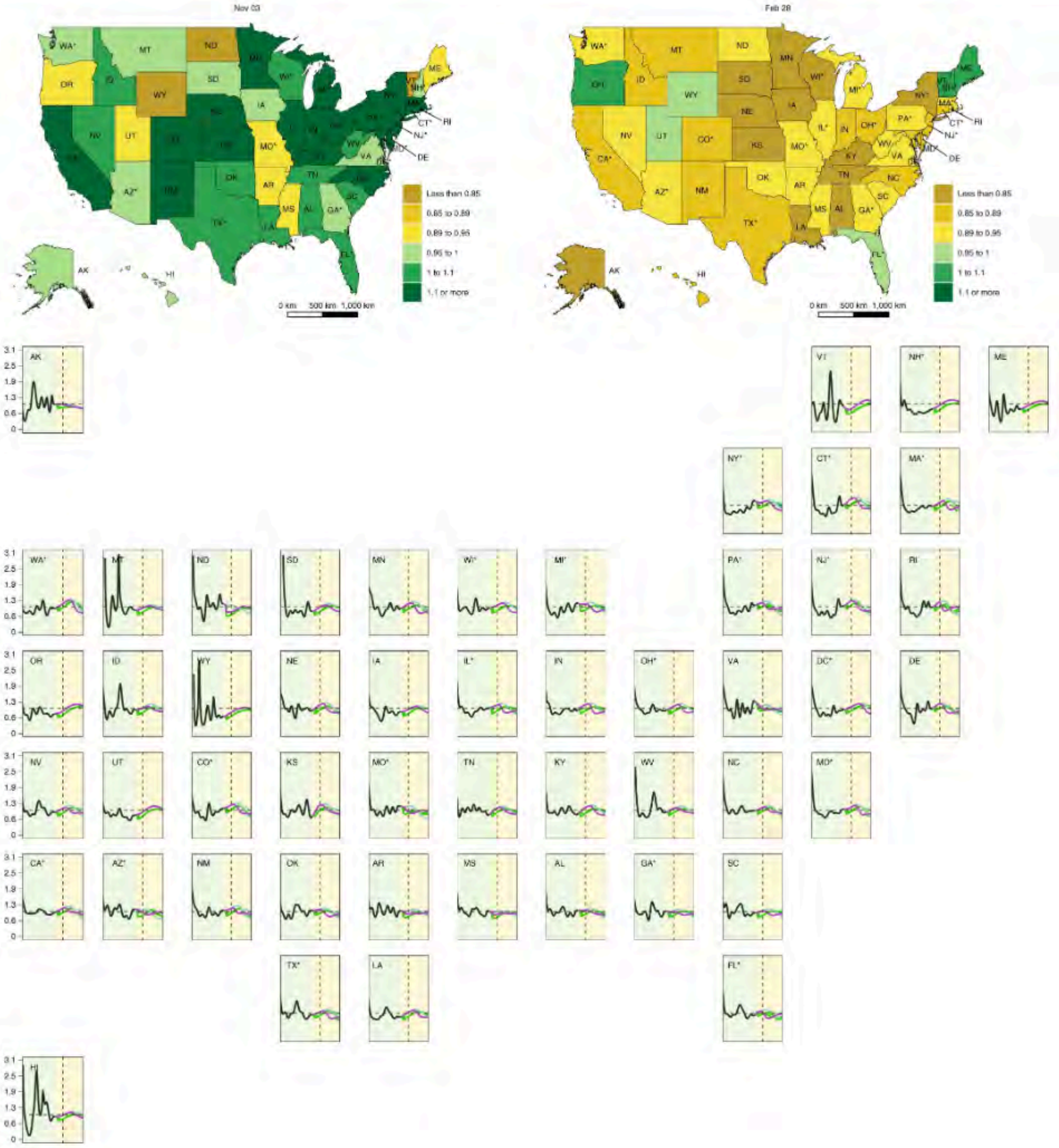
Services provided by Planned Parenthood
2,500,000
2,000,000
1,500,000
1,000,000
500,000
0
2006
2,007,371
Cancer screening and
prevention services
2013
935,573
2006
289,750
Abortions
2013
327,000
2006
2013
SOURCE: Planned Parenthood
Vox

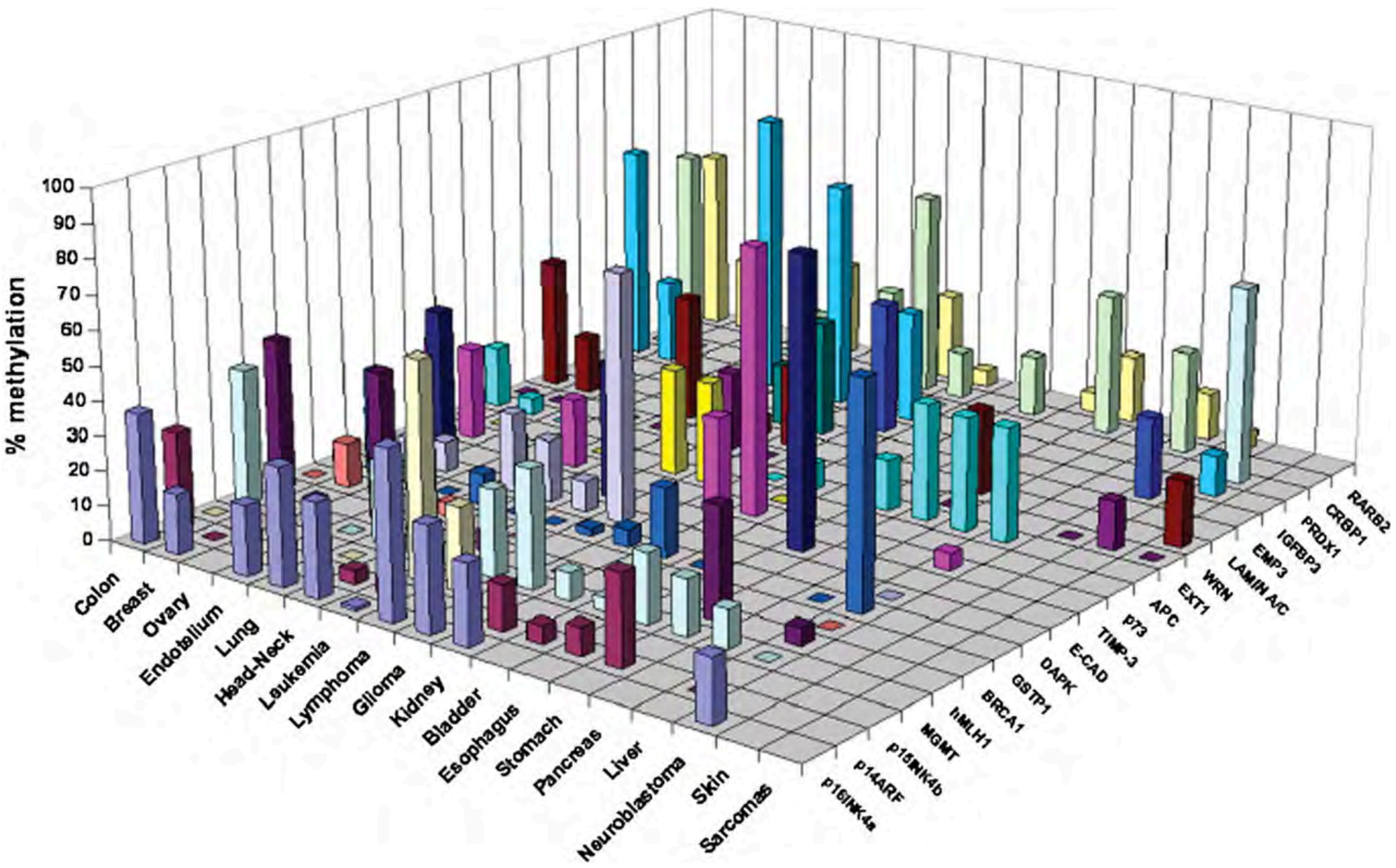

a
COL
OLA
CPT
HU
BTZ
DDR/RSR
chromatin remodeling
DNA replication
HR
ICL
MMEJ
NHEJ
SSBR/BER
Pathway
Z-score
5.0
2.5
0.0
−2.5
−5.0
- log FDR
10
20



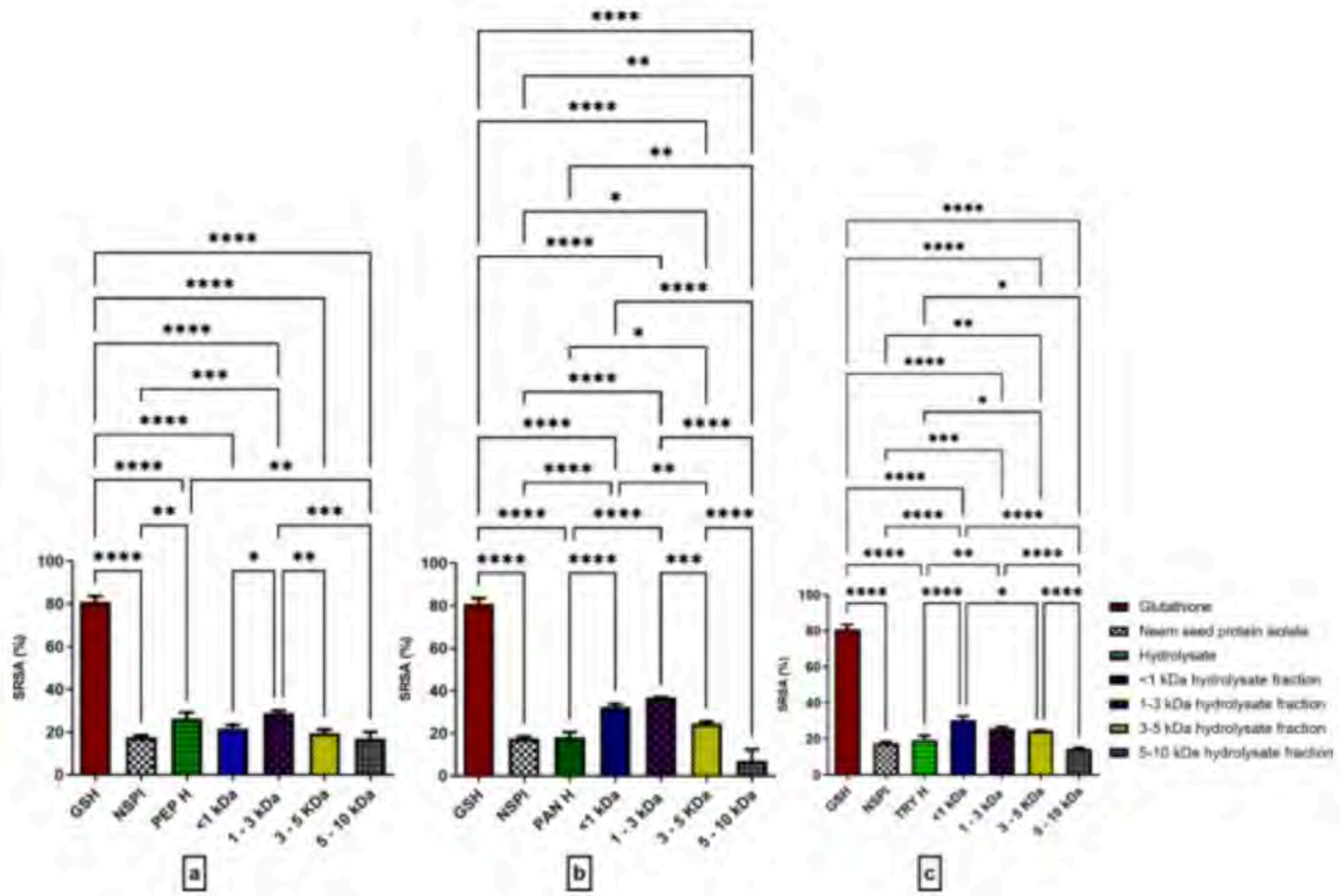
PCoA 2 [4.4%]
0.2
0.0
-0.2
-0.2
0.0
0.2
PCoA 1 [8.9%]
Breeding status
Breeding
Non-breeding
Sex/Age
Male
Female
Juvenile
PCoA 2 [14.4%]
0.2
0.0
-0.2
-0.4
-0.4
-0.2
0.0
0.2
PCoA 1 [21.5%]
Breeding status
Breeding
Non-breeding
Sex/Age
Male
Female
Juvenile


Distribution of All TFBS Regions
Pseudogene/ ambiguous 17%
5' to known gene 22%
Novel 24%
Within or 3' flanking to a known gene 36%
866 Total TFBS Regions


Nov 03
Feb 28
Less than 0.85
0.85 to 0.89
0.89 to 0.95
0.95 to 1
1 to 1.1
1.1 or more
0 km 500 km 1,000 km
3.1
2.5
1.9
1.3
0.6
0



% methylation
100
90
80
70
60
50
40
30
20
10
0
Colon
Breast
Ovary
Endotelium
Lung
Head-Neck
Leukemia
Lymphoma
Glioma
Kidney
Bladder
Esophagus
Stomach
Pancreas
Liver
Neuroblastoma
Skin
Sarcomas
p16INK4a
p14ARF
p15INK4b
MGMT
hMLH1
BRCA1
GSTP1
DAPK
E-CAD
TIMP-3
p73
APC
EXT1
WRN
LAMIN A/C
EMP3
IGFBP3
PRDX1
CRBP1
RARB2


SRSA (%)
GSH
NSPI
PEP H
PAN H
TRY H
<1 kDa
1 - 3 kDa
3 - 5 kDa
5 - 10 kDa
a
b
c
Glutathione
Neem seed protein isolate
Hydrolysate
<1 kDa hydrolysate fraction
1-3 kDa hydrolysate fraction
3-5 kDa hydrolysate fraction
5-10 kDa hydrolysate fraction

Recommended resources

Books:

Tufte, E. 2001. *The Visual Display of Quantitative Information*. Graphics Press. (2nd ed.)

Wilke, C. 2019. *Fundamentals of Data Visualization*. O'Reilly Media.

Wickham, H. 2025. *ggplot2: Elegant Graphics for Data Analysis*. Springer. (3rd ed.)

Websites:

From Data to Viz: https://www.data-to-viz.com/

FriendsDontLetFriends: https://github.com/cxli233/FriendsDontLetFriends

Beautiful Plotting in R: https://www.cedricscherer.com/2019/08/05/a-ggplot2-tutorial-for-beautiful-plotting-in-r/

R Graph Gallery: http://rgraphgallery.blogspot.com/

MatPlotLib Examples: https://matplotlib.org/stable/plot_types/index.html

Seaborn Gallery: https://seaborn.pydata.org/examples/index.html